

北京師範大學

硕士学位论文

Master Dissertation

论文题目: **Quantifying Asset Specificity in Economic Transactions: A Machine Learning Approach**

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学科专业 (Major): World Economy

完成日期 (Completion Date): March, 2024

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Provost's Office and Academic Affairs (Graduate School)
Beijing Normal University

Beijing Normal University Academic Dissertation

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Quantifying Asset Specificity in Economic Transactions: A Machine Learning Approach

ABSTRACT

This thesis investigates asset specificity within economic transactions, utilizing advanced machine learning (ML) techniques and causal inference methods to provide a quantitative analysis and categorization of asset specificity levels. Positioned at the intersection of machine learning and economic theory, particularly transaction cost economics, this study addresses the critical challenge of measuring and characterizing asset specificity—a concept fundamental to understanding the extent to which assets are tailored for specific transactions and their redeploy ability. Leveraging a comprehensive dataset encompassing various financial and operational features, the research develops predictive models to classify assets into low, medium, and high specificity categories. Employing Random Forest classification, a robust ML algorithm, the study achieves significant predictive accuracy, demonstrating the efficacy of ML techniques in enhancing empirical economic analysis. The findings reveal that financial magnitude, operational timelines, and the nature of participating entities significantly influence asset specificity, with the 'frequency of transactions' emerging as the most critical predictor. To deepen the understanding of asset specificity, Causal ML and Structural Equation Modeling (SEM) analyses were employed. Causal ML, particularly uplift modeling, was used to estimate the causal impact of financial investments and transaction frequency on asset specificity. The analysis revealed that higher financial investments correlate with increased asset specificity, and banks have a positive influence on high specificity but a negative impact on low specificity. R&D missions were linked to higher specificity, while export activities exhibited variable effects.

The SEM analysis highlighted the critical role of transaction frequency, especially involving banks, in influencing both high and low specificity. It showed that monetary investments have minimal direct effects but complex indirect relationships, particularly in reducing patent outcomes. The duration of transactions was found to be insignificant, whereas foreign investments significantly impacted specificity but were less likely when banks were involved. Patent outcomes consistently showed a negative impact on specificity, indicating a trend towards more general innovations, while R&D missions were associated with higher specificity. This integrated approach not only bridges the theoretical and empirical realms of transaction cost economics but also offers actionable insights for optimizing asset management and contractual strategies. The study underscores the value of combining data-driven ML approaches with economic theory,

suggesting a paradigm shift towards more empirical and quantitative analyses in economics.

KEY WORDS: Asset Specificity, Machine Learning, Economic Transactions, Uplift Modeling, CausalML, Sequential Modeling, Transaction Cost Economics, Random Forest Classification, Predictive Modeling.

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Chapter 1 Introduction

1.1 Background and Motivation

In my thesis, I explore a novel approach to quantifying asset specificity within the domain of business asset management. Asset specificity refers to the degree to which an asset is tailored to the unique needs of a business, thereby impacting its interchangeability and versatility for other uses. Traditional methods rely on qualitative assessment, examining contractual details, repair costs, and expert opinions to determine this specificity. My work innovates by introducing a quantifiable measure using a machine learning algorithm, Random Forest. This method analogously considers various predictive factors, similar to gathering diverse weather opinions for camping preparation, to arrive at a consensus. The algorithm processes extensive datasets, including transactional frequency, financial details, contract duration, and intellectual property presence. My findings highlight that transaction frequency, monetary value, contract length, buyer identity (e.g., whether a bank), and intellectual property inclusion are significant in determining asset specificity. Machine learning (ML) methods have transformed many fields by offering advanced tools for predicting outcomes, recognizing patterns, and making data-driven decisions. For our study on asset specificity, these methods are particularly useful.

Machine learning algorithms, such as random forests and gradient boosting machines, are excellent at predictive modeling because they can identify complex, non-linear relationships in the data. In our study, we used these methods to predict the level of asset specificity based on various financial, transactional, and contextual features. Unlike traditional statistical methods, which might miss these complex patterns, machine learning models can leverage large datasets to make accurate predictions.

Understanding which factors most influence asset specificity is crucial for making strategic decisions. Machine learning techniques, like random forests, help us understand the importance of different features. For example, our analysis showed that financial metrics like 'money' and 'foreign investment' significantly impact asset specificity. This information helps managers prioritize their investments and optimize their asset management strategies.

Machine learning methods, especially those focused on causal inference (like uplift modeling), allow us to estimate the causal impact of specific interventions or features on asset specificity. This is essential for understanding how changes in financial investment or transaction frequency might affect asset specificity levels. By identifying causal relationships, our study provides actionable insights for decision-makers.

Traditional econometric models, such as linear regression, often struggle with complex, non-linear relationships and high-dimensional data. These models typically rely on assumptions about the data that may not hold true in real-world scenarios. For example,

they may assume a linear relationship between predictors and the outcome, which can oversimplify the reality and lead to less accurate predictions. Imagine trying to predict asset specificity using linear regression: if the relationship between financial investment and asset specificity is non-linear (e.g., higher investments initially increase specificity, but after a certain point, the effect plateaus or even decreases), a linear model would fail to capture this complexity. Another example is the relationship between transaction frequency and asset specificity. If transaction frequency initially has a positive impact on specificity due to increased specialization, but beyond a certain frequency, the effect diminishes or becomes negative due to diminishing returns or increased operational complexity, a linear model would miss this nuanced behavior. Additionally, considering the scenario where multiple factors interact with each other to influence asset specificity, for instance, the combined effect of the type of buyer (e.g., bank vs. non-bank) and the industry of the seller (e.g., tech vs. non-tech) on asset specificity might be significant. Traditional models may struggle to account for these interactions without explicitly including interaction terms, which can quickly become unmanageable with many variables. In contrast, machine learning models like random forests can naturally handle such interactions by splitting data based on different combinations of features, thereby uncovering intricate patterns without requiring manual specification of interaction terms.

Traditional econometric analysis typically involves writing equations that describe the relationships between variables. These equations are based on predefined assumptions that are necessary for the validity and interpretability of the models. For example, a simple linear regression model might be written as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

Here, Y is the dependant variable (the outcome we are trying to predict). $X_1, X_2, \dots, X_n$ are the independent variables. β_0 is the intercept, $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients that measure the impact of the predictors on Y , and ϵ is the error term that captures the variation in Y not explained by the predictors. Traditional econometric models rely on several key assumptions to ensure their validity. Example: If we assume that the effect of financial investment on asset specificity is linear, we would write an equation like,

Asset Specificity: $\beta_0 + \beta_1 \times \text{Investment} + \epsilon$

However, this may not capture non-linear effects, such as diminishing returns on investment. The observations are independently and identically distributed. Example: Each transaction's asset specificity is assumed to be independent of others. This assumption can be problematic if there are correlations within the data, such as repeated

transactions by the same entity. The variability in asset specificity should be consistent across different levels of financial investment. If high investments lead to greater variability in specificity, this assumption is violated. If 'financial investment' and 'foreign investment' are highly correlated, it becomes difficult to isolate their individual effects on asset specificity. The deviations from the predicted asset specificity should follow a normal distribution. If the error terms are skewed, it can affect the reliability of hypothesis tests.

These assumptions are crucial because they underpin the statistical properties of the estimators used in econometric models. For instance, the Ordinary Least Squares (OLS) estimator is unbiased and efficient (i.e., it has the smallest variance among all unbiased estimators) only if these assumptions hold. In contrast, machine learning models do not always require the specification of an explicit equation or adherence to strict assumptions about the data. Instead, machine learning algorithms are designed to learn patterns directly from the data. For example:

Random Forests: These models build multiple decision trees and aggregate their predictions. They can capture non-linear relationships and interactions between variables without requiring the user to specify these relationships explicitly.

Gradient Boosting Machines: These models iteratively build a series of weak learners, typically decision trees, to minimize prediction errors. They can handle complex data structures and high-dimensional spaces.

Machine learning models are often more flexible than traditional econometric models, allowing them to capture intricate patterns in the data that econometric models might miss due to their reliance on predefined equations and assumptions. While traditional econometric models are valuable for their interpretability and well-established theoretical foundations, machine learning methods offer more flexibility and accuracy in complex, high-dimensional settings. By applying machine learning to our study of asset specificity, we can uncover deeper insights and make more informed decisions, ultimately leading to better management and optimization of assets.

Ultimately, my aim is to contribute to the field of transaction cost economics by integrating machine learning methods. This should enhance our theoretical understanding while also demonstrating practical applications in addressing real-world challenges related to asset specificity.

1.2 Problem Statement

The problem addressed in this thesis revolves around the need to accurately measure and characterize asset specificity within the realm of economic transactions. Asset specificity refers to the degree to which investments made for particular transactions cannot be easily transferred or redeployed without incurring significant losses. Despite its fundamental importance in transaction cost economics, there exists a notable gap in empirical studies dedicated solely to understanding and quantifying asset specificity. This gap presents several challenges. Firstly, the lack of comprehensive empirical analysis hinders our ability to fully grasp the nuanced nature of asset specificity and its implications for economic decision-making. Without robust measurement techniques, businesses and policymakers may struggle to assess the true costs and risks associated with specific transactions, leading to suboptimal contract management and policy formulation. The traditional methods of economic analysis often fall short in capturing the multifaceted aspects of asset specificity. As contractual agreements become increasingly intricate, there is a pressing need for more sophisticated analytical tools to accurately evaluate asset specificity and its impact on various economic outcomes.

1.3 Research Objective

The primary objective of this research is to innovate at the intersection of machine learning (ML) and economic theory, specifically focusing on the concept of asset specificity within transaction cost economics. This thesis aims to leverage the capabilities of machine learning classification algorithms to systematically categorize assets based on their specificity levels, thus providing a nuanced understanding of their strategic importance to firms. By doing so, it seeks to bridge the gap between the theoretical underpinnings of asset specificity and the practical, data-driven analysis of economic transactions. The detailed objectives of this research are as follows:

To Develop a Predictive Model: Design, train, and validate a machine learning classification model that can accurately predict the level of asset specificity in economic transactions. This involves selecting appropriate ML algorithms, processing and preparing the dataset for analysis, and evaluating the model's performance through rigorous testing.

To Quantify Asset Specificity: Translate the qualitative aspects of asset specificity into quantifiable measures that can be analyzed through machine learning. This includes developing a methodology for binning or categorizing transaction data into distinct levels of asset specificity, such as low, medium, and high, based on financial and other relevant indicators.

To Analyze Influential Factors: Identify and quantify the factors that significantly influence asset specificity in transactions. This entails utilizing the feature importance output of machine learning models to understand which variables (e.g., transaction

frequency of transactions, type of buyer and seller, temporal characteristics) play a pivotal role in determining asset specificity.

To Evaluate the Practical Implications: Investigate the practical implications of the machine learning model's predictions for firms and policymakers. This includes understanding how the findings can aid in strategic planning, risk management, and decision-making regarding asset investments and contract negotiations.

To Contribute to Economic Theory: Enhance the theoretical discourse on asset specificity by providing empirical evidence and data-driven insights. This research aims to contribute to the field of transaction cost economics by offering a novel approach to evaluating asset specificity that complements traditional qualitative assessments.

To Foster Interdisciplinary Research: Bridge the gap between machine learning and economic theory, demonstrating the potential of interdisciplinary approaches to address complex economic questions. This research seeks to inspire further studies that leverage machine learning for economic analysis, encouraging a fusion of technical and theoretical expertise.

Through the achievement of these objectives, this thesis aspires to make a significant contribution to the understanding of asset specificity, offering a fresh perspective that combines economic theory with the power of machine learning. It is anticipated that the insights gained will not only enrich academic discourse but also provide tangible tools and methodologies that can transform the landscape of transaction analysis and asset management in practice.

1.4 Thesis Structure

The "Thesis Structure" section of this document delineates the organized framework through which the research unfolds, guiding the reader through the logical progression of the study from its inception to its conclusions. This structured approach is designed to facilitate a comprehensive understanding of the research objectives, methodology, findings, and their broader implications. Here is a detailed breakdown of the thesis structure:

Introduction: This opening chapter sets the stage for the entire study, presenting the background, problem statement, and motivation behind the research. It outlines the central questions that the thesis aims to answer and introduces the key concepts of asset specificity and machine learning within the context of economic analysis.

Literature Review: The second chapter delves into a thorough review of existing literature, covering foundational theories related to asset specificity, contract outcomes, and the application of machine learning in economic research. This section establishes

the theoretical underpinnings of the study and highlights gaps in the current body of knowledge that the research aims to address.

Methodology: Here, the thesis details the research methodology, including data collection processes, the selection of machine learning algorithms, and the analytical techniques employed to examine the impact of asset specificity on contract outcomes. This chapter is crucial for understanding the empirical approach taken and the rationale behind the choice of methods.

Data Analysis: This chapter presents the core empirical analysis conducted as part of the research. It includes a comprehensive examination of the dataset, the preprocessing steps, the training and validation of the machine learning models, and the analysis of the results. The section aims to provide transparency into the analytical process and the derivation of insights from the data.

Results: The fifth chapter outlines the findings of the study, focusing on the predictive accuracy of the machine learning models in determining asset specificity. This section discusses the implications of the results in the context of the research questions and the broader theoretical framework. Additionally, it incorporates insights from CausalML and Structural Equation Modeling (SEM) analyses, providing a comprehensive understanding of the factors influencing asset specificity. By integrating these advanced analytical methods, the chapter highlights both the predictive power of machine learning and the nuanced causal relationships uncovered through SEM, offering a robust and multifaceted perspective on asset specificity.

Discussion: This chapter interprets the findings, integrating them with existing literature and theoretical concepts. It explores the practical implications of the research for businesses and policymakers, addresses the limitations of the study, and suggests areas for future research. The discussion aims to contextualize the results within the larger discourse on economics and machine learning.

Conclusion: The final chapter summarizes the key contributions of the thesis, reiterates the significance of the findings, and reflects on the broader impact of the research. It provides a concise overview of the study's achievements and outlines the potential for future work stemming from this research.

References: This section lists all the scholarly works cited throughout the thesis, providing a comprehensive bibliography that supports the research.

Chapter 2 Literature Review

2.1 Comparison with Existing Literature

This section compares the findings of the current study with existing literature on asset specificity, transaction cost economics, and the application of machine learning in economic analysis. By situating our results within the broader academic discourse, we can discern both the convergence and divergence from previous studies, thereby highlighting the unique contributions and potential avenues for further research. The concept of asset specificity, as introduced by Williamson (1985) in the context of transaction cost economics, posits that transactions requiring specific assets have profound implications for governance structures and market dynamics. Prior literature has predominantly explored asset specificity through qualitative analyses or econometric models, focusing on its impact on contractual arrangements and organizational forms. Our study aligns with these foundational principles, confirming the significant role of financial metrics in determining asset specificity. However, by applying machine learning classification, this thesis extends the traditional analysis, offering a quantified and predictive perspective on asset specificity that has been less emphasized in existing literature. Recent advances in data science have spurred interest in applying machine learning techniques to economic research. Studies such as Varian (2014) have advocated for the potential of machine learning to uncover patterns and relationships in economic data beyond the reach of traditional statistical methods. While the application of machine learning in economics is burgeoning, its use in analyzing transaction-specific attributes, particularly asset specificity, remains underexplored. Our study contributes to this emerging field by demonstrating how machine learning can not only complement traditional economic analyses but also provide novel insights into complex economic phenomena like asset specificity.

Empirical investigations into asset specificity have varied in their methodological approaches and findings. Some studies, such as Joskow (1987), have examined the implications of asset specificity in specific industries, such as electricity generation, through case studies and industry analyses. Others have employed econometric models to assess the impact of asset specificity on firm performance and contractual outcomes. The current study diverges from these approaches by utilizing a machine learning model to classify transactions based on their level of asset specificity, leveraging a comprehensive dataset from government database. This approach not only corroborates the theoretical importance of financial magnitude in asset specificity but also provides a scalable and objective method for assessing asset specificity across diverse transaction contexts. Teng, Li, and Chang (2020) delve into the incorporation of machine learning in empirical asset pricing models, highlighting neural networks' superiority over traditional models in predicting stock returns. This study exemplifies the potential of machine learning to provide more accurate financial forecasting and risk assessment (Teng, Li, & Chang, 2020). Mocnik (2001) explores how asset specificity influences a

firm's ability to borrow, linking the nature of a firm's assets to its capital structure and risk profile. The work underscores the relevance of asset specificity in determining the allocation of risk and control among investors (Mocnik, 2001). Jati et al. (2021) examine the impact of asset specificity on financial strategy and conditional conservatism. Their findings contribute to a nuanced understanding of how the unique attributes of asset specificity shape financial reporting and market expectations (Jati, Naufal, Arisanti, & Syam, 2021). The effects of asset specificity on maintenance financial performance, especially within the healthcare industry, are analyzed by Cruz et al. (2014). This research offers valuable insights into how asset specificity can affect the cost-effectiveness and efficiency of service delivery in sectors characterized by high-value investments (Cruz, Haugan, & Rincón, 2014). Lin, Wu, and Chiou (2017) investigate how asset specificity can be leveraged by suppliers to increase customer dependence. Their findings reveal the strategic use of asset specificity to maintain and solidify customer-supplier relationships (Lin, Wu, & Chiou, 2017). Delbufalo (2021) presents a meta-analysis of asset specificity's direct effects on relationship performance. This study harmonizes theoretical tensions between Transaction Cost Economics and Relational Exchange Theory, offering a more comprehensive view of the role of asset specificity in buyer-supplier relationships (Delbufalo, 2021). The overarching role of machine learning in asset management is provided by Lee et al. (2023), who offer an overview of various machine learning models and their application in asset management, presenting both the opportunities and challenges this technology brings to the financial sector (Lee, Thompson, Kim, Kim, & Fabozzi, 2023). Similarly, Weigand (2019) reviews the use of machine learning in empirical asset pricing, emphasizing the necessity for critical consideration of these methodologies due to their nascent application in the finance domain (Weigand, 2019). The theoretical underpinnings of asset specificity are critically reviewed by De Vita, Tekaya, and Wang (2011), who dissect its definitional complexities and multifaceted nature as reflected in the empirical literature. They emphasize the need for clarity and consistency in its operationalization for future research (De Vita, Tekaya, & Wang, 2011). Expanding the scope of machine learning methods in asset management, Mirete-Ferrer et al. (2022) critically analyze the current state-of-the-art and lay down future research directions, highlighting the significant contributions and gaps in the field (Mirete-Ferrer, Garcia-Garcia, Baixauli-Soler, & Prats, 2022). Chun-hai's (2005) theoretical review discusses the applications of asset specificity across economic areas, providing a foundational analysis helpful for economic research (Chun-hai, 2005). Lastly, Lamminmaki (2005) applies Williamson's framework to appraise outsourcing activities in the hotel industry, exploring how asset specificity manifests in this specific context (Lamminmaki, 2005). These diverse explorations into the dimensions of asset specificity demonstrate its centrality in economic thought and practice. The integration of machine learning promises to open new avenues for the application, measurement, and theoretical understanding of asset specificity, offering a transformative lens through which future economic research and financial strategy can be viewed. A comprehensive summary of existing literature viewpoints:

1. Traditional Approaches:

Williamson (1985): Introduced the concept of asset specificity within the framework of transaction cost economics, emphasizing its impact on governance structures and market dynamics.

Joskow (1987): Explored asset specificity in the electricity generation industry through case studies and industry analyses, highlighting the importance of specific investments in long-term contractual arrangements.

Mocnik (2001): Investigated the influence of asset specificity on a firm's ability to borrow, linking asset characteristics to capital structure and risk profiles.

2. Econometric Models:

De Vita, Tekaya, and Wang (2011): Critically reviewed the definitional complexities of asset specificity and its empirical operationalization, calling for clarity and consistency in measurement.

Jati et al. (2021): Examined the impact of asset specificity on financial strategy and conditional conservatism using econometric models, contributing to a nuanced understanding of its role in financial reporting and market expectations.

3. Recent Methodological Advances:

Delbufalo (2021): Presented a meta-analysis harmonizing theoretical tensions between Transaction Cost Economics and Relational Exchange Theory, providing a comprehensive view of asset specificity in buyer-supplier relationships.

Lin, Wu, and Chiou (2017): Investigated how suppliers leverage asset specificity to increase customer dependence, using empirical data to demonstrate strategic uses in maintaining customer relationships.

4. Contractual and Organizational Implications:

Williamson (1985): Highlighted the role of asset specificity in shaping governance structures, with high specificity necessitating complex contractual safeguards to mitigate opportunism.

Joskow (1987): Demonstrated that high asset specificity in the electricity sector leads to long-term contracts to secure investments.

5. Financial and Strategic Implications:

Mocnik (2001): Linked asset specificity to firm borrowing capacity, influencing capital structure and investor risk allocation.

Cruz et al. (2014): Analyzed the effects of asset specificity on maintenance and financial performance in the healthcare industry, highlighting cost-effectiveness and efficiency impacts.

6. Market Dynamics and Strategic Management:

Jati et al. (2021): Showed that asset specificity influences financial strategy and market expectations, with significant implications for conditional conservatism in financial reporting.

Lin, Wu, and Chiou (2017): Emphasized the strategic use of asset specificity by suppliers to secure and maintain customer relationships.

7. Integration of Machine Learning:

Teng, Li, and Chang (2020): Highlighted the superiority of machine learning models, such as neural networks, in predicting financial outcomes compared to traditional models. This study underscores the potential of machine learning in providing more accurate financial forecasting and risk assessment.

Weigand (2019): Reviewed the use of machine learning in empirical asset pricing, emphasizing the need for critical evaluation due to the nascent application of these methods in finance.

Varian (2014): Advocated for the use of machine learning to uncover patterns and relationships in economic data, which traditional statistical methods might miss. However, the application of machine learning to asset specificity is still underexplored.

Lee et al. (2023): Provided an overview of machine learning models in asset management, discussing the opportunities and challenges these technologies bring to the financial sector.

Mirete-Ferrer et al. (2022): Critically analyzed the state-of-the-art in machine learning methods for asset management, highlighting significant contributions and gaps. This study lays down future research directions for integrating machine learning in asset specificity analysis.

Chun-hai (2005): Discussed theoretical applications of asset specificity across various economic areas, providing foundational analysis helpful for integrating machine learning in future research.

2.2 Gap in Literature

The integration of machine learning techniques to assess asset specificity in economic transactions illuminates a significant gap in the existing literature. While transaction cost economics, pioneered by Oliver Williamson, provides a robust framework for understanding the implications of asset specificity on organizational behavior and market structure, the application of quantitative models, particularly machine learning, to explore these concepts is markedly underrepresented. This section delineates the gap identified in the literature, positioning the current study within the broader academic discourse.

Transaction cost economics has traditionally relied on qualitative analyses and case studies to elucidate the role of asset specificity in economic transactions. The seminal works by Williamson (1979, 1985) underscore the importance of asset specificity in determining the optimal governance structure for transactions, suggesting that higher levels of specificity increase the reliance on hierarchical

arrangements as opposed to market-based transactions. While these theoretical underpinnings are well-established, the literature exhibits a paucity of empirical studies employing advanced statistical or machine learning methods to quantify asset specificity and its impacts.

Most studies on asset specificity have mainly relied on econometric models to understand how it affects organizational behavior and market dynamics. These studies have been instrumental in validating theories, demonstrating how high asset specificity can increase transaction costs and foster vertical integration within supply chains. However, asset specificity is inherently complex and multifaceted, which poses challenges for traditional econometric methods to fully capture. These methods often rely on simplifying assumptions and struggle to accurately depict the intricate dimensions of specificity.

The application of machine learning in economics has been growing, focusing mainly on predictive modeling and policy evaluation. Nonetheless, its use in exploring foundational economic theories, particularly transaction cost economics, remains limited. The existing literature has not fully leveraged machine learning's potential to uncover hidden patterns in large datasets or to provide a nuanced, quantitative analysis of theoretical constructs such as asset specificity. By employing machine learning to quantify asset specificity, this study not only contributes to the empirical literature on transaction cost economics but also expands the methodological toolkit available to economists. Our study demonstrates the feasibility and value of using machine learning for theoretical exploration and hypothesis testing, paving the way for future research to apply similar techniques across a range of economic theories and concepts. By quantifying asset specificity through machine learning models, we provide a more objective and scalable method for its assessment, addressing the limitations of traditional econometric approaches. This integration offers new insights into how asset specificity impacts transaction costs and governance structures, ultimately enriching the theoretical and empirical understanding of transaction cost economics.

2.3 Contribution of this Thesis

Using a machine learning approach, specifically a random forest algorithm, to determine the feature importance scores in this thesis adds a modern analytical dimension to the study of asset specificity that diverges from Williamson's original theoretical framework. Williamson's transaction cost economics (TCE) is primarily theoretical, offering a conceptual understanding of how transactions, asset specificity, and governance structures interact. By employing a random forest algorithm, I have added an empirical layer that not only assesses the importance of various features but also provides predictive insights into how these features may influence asset specificity and transaction outcomes in real-world scenarios. This quantitative method allows for the identification and ranking of factors based on their influence on the model's

outcome, a direct application of theory to data not explicitly covered in Williamson's work.

Random forest algorithms are capable of capturing non-linear relationships between variables, which means my approach can uncover complex interactions between factors affecting asset specificity that traditional linear models or theoretical frameworks might miss. This capability enhances the understanding of asset specificity by revealing intricate dynamics that influence transaction costs and governance decisions beyond Williamson's original scope. While Williamson's TCE framework emphasizes the significance of asset specificity, uncertainty, and frequency of transactions, my use of a random forest algorithm quantitatively identifies the relative importance of a wide range of features. This data-driven approach provides a nuanced view of asset specificity, highlighting specific factors like the frequency of transactions, the role of money, time, and the nature of the investing entities as critical determinants. It goes beyond the binary market versus hierarchy decision to illustrate a spectrum of influences that drive governance structure choices. My thesis reflects an adaptation of TCE principles to contemporary economic contexts, incorporating variables like foreign investment, intellectual property outcomes, and the specifics of the buyer-seller relationship. The random forest's ability to handle high-dimensional data means i can include a diverse set of variables reflecting today's complex economic transactions, offering insights that are directly relevant to current business practices. Finally, the objective ranking of features based on their importance scores provides a clear hierarchy of factors influencing asset specificity. This empirical evidence complements Williamson's theoretical contributions by specifying which factors are most critical in real-world datasets, potentially guiding future theoretical and empirical research in economics and beyond. By integrating machine learning with economic theory, this thesis represents an innovative approach that combines the strengths of both to offer deeper insights into the determinants of asset specificity and their implications for governance structures in the modern economy.

In addition to the random forest analysis, I employed causal ML techniques to further investigate the causal relationships between different variables and asset specificity. Causal ML, which includes methods like uplift modeling, allows for a more nuanced understanding of how specific interventions or changes in one variable may causally affect asset specificity. This approach provides robust evidence of causality rather than mere correlation, thereby strengthening the validity of the findings and offering actionable insights for decision-makers in economic and business contexts.

Chapter 3 Theoretical Framework On Asset Specificity

3.1 Concept of Asset Specificity

Asset specificity is a key concept in transaction cost economics, which highlights the extent to which assets are customized or tailored for a particular transaction or relationship. This concept plays a crucial role in understanding how economic actors design contracts and structure their relationships to minimize transaction costs and safeguard investments. Asset specificity arises when an asset is designed to serve a specific transaction with one or a few parties, making it less valuable or even useless outside this specific context. The degree of specificity can significantly affect the parties' bargaining power, influence their behavior due to the mutual dependencies created, and necessitate mechanisms to safeguard against opportunistic actions. Understanding asset specificity is vital for analyzing how specific investments influence the governance structures of transactions, including the choice between market, hybrid, or hierarchical forms of organization.

3.2 Types of Asset Specificity

Asset specificity can be categorized into several types, each with distinct characteristics and implications for contractual relationships and organizational governance:

Physical Asset Specificity: Refers to assets that are physically unique or tailored for specific transactions, such as specialized machinery or equipment designed for a particular production process.

Human Asset Specificity: Involves skills, knowledge, or expertise that employees acquire, which are specific to the firm's operations or to the execution of a specific contract.

Site Specificity: Pertains to the geographical location of assets, making proximity a critical factor for the execution of certain transactions, such as the location of a supplier relative to a buyer.

Dedicated Assets: These are investments made in anticipation of a particular transaction or series of transactions with a specific party, such as additional production capacity to meet the demands of a particular customer.

Temporal Specificity: Relates to the timing of transactions, where the value of an asset is significantly affected by specific time frames or deadlines.

Understanding these types helps in analyzing how the specificity of assets affects transaction costs and the design of contracts and governance structures.

3.3 Implications of Asset Specificity on Contracts

The implications of asset specificity on contracts are profound, shaping how parties manage risk, structure agreements, and resolve disputes. High asset specificity often leads to a greater risk of opportunistic behavior, as the party investing in the specific asset may become vulnerable to hold-up by the other party, who might exploit the situation to renegotiate terms. To mitigate these risks, contracts involving specific assets typically incorporate detailed provisions, including:

Safeguards and Guarantees: These may include performance bonds, warranties, and penalties for non-compliance, designed to protect the interests of the party making the specific investment.

Flexible Contracting: Given the difficulty of anticipating all future contingencies in highly specific transactions, contracts may include clauses that allow for renegotiation or adaptive governance mechanisms.

Relational Norms: Especially in long-term relationships involving specific assets, parties often rely on relational norms and mutual trust, beyond formal contract terms, to govern their interactions.

Vertical Integration: In cases of extreme asset specificity, firms may opt for vertical integration, owning the assets and operations along the supply chain, to control the transaction and reduce dependency on external parties.

Understanding the implications of asset specificity is crucial for firms as they navigate the complexities of economic transactions, ensuring that investments are protected and that the potential for opportunistic behavior is minimized.

Chapter 4 Machine Learning In Economics and Contract Analysis

4.1 Introduction to Machine Learning

Machine Learning (ML) is a branch of artificial intelligence (AI) that focuses on building algorithms and statistical models that enable computers to perform specific tasks without using explicit instructions, relying instead on patterns and inference. It is a method of data analysis that automates analytical model building, allowing machines to adapt to new scenarios independently. The core principle of machine learning is to learn from data, identify patterns, and make decisions with minimal human intervention. The concept of machine learning has its roots in the early days of computer science and artificial intelligence. The idea that machines could learn and adapt their behaviors without being explicitly programmed to perform specific tasks was revolutionary. Early examples include Arthur Samuel's checkers-playing program in the 1950s and the development of neural networks in the 1960s. Over the decades, advances in computing power, data storage, and algorithmic complexity have propelled machine learning into a key technology that underpins many modern applications.

Machine learning algorithms are generally classified into three main types based on their learning style: supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning: This type involves learning a function that maps an input to an output based on example input-output pairs. It infers a function from labeled training data, consisting of a set of training examples.

Unsupervised Learning: In unsupervised learning, the algorithm learns patterns from untagged data. The system tries to learn without a teacher, identifying commonalities in the data and reacting based on the presence or absence of such commonalities in each new piece of data.

Reinforcement Learning: This type is about taking suitable action to maximize reward in a particular situation. It is employed by various software and machines to find the best possible behavior or path it should take in a specific situation.

Machine learning is ubiquitous today, with applications ranging from the mundane to the sophisticated. It powers web search engines, real-time ads on web pages, email filtering, recommendation systems, and much more. Beyond consumer applications, ML is used in health care for predictive diagnostics, in finance for credit scoring and algorithmic trading, in manufacturing for predictive maintenance, and in many other industries.

While machine learning has achieved remarkable successes, it also faces significant challenges. These include issues related to data privacy, security, algorithmic bias, and the interpretability of machine learning models. Moreover, as machine learning continues to evolve, there is a growing need for models that can learn more efficiently from smaller data sets, adapt in dynamic environments, and explain their decision-making processes in understandable terms. The future of machine learning is closely tied to advancements in AI, data science, and computational hardware. As these fields continue to evolve, we can expect machine learning to become even more integral to technological innovation and our daily lives, pushing the boundaries of what computers can achieve. Machine learning represents a pivotal advancement in the way computers process information, offering the potential to transform industries and enhance human capabilities. As we continue to explore the possibilities of this technology, it is imperative to address its challenges and ensure it serves the broader interests of society.

4.2 Rationale for Employing Machine Learning in the Thesis

The decision to employ machine learning (ML) methodologies in this thesis is grounded in several key considerations, each aimed at enhancing the investigation into asset specificity and its implications for economic transactions. The rationale behind this choice encompasses the need for advanced analytical capabilities, the complexity of economic data, and the goal of pioneering methodological innovations in the field of economics. Below are detailed justifications for incorporating ML into this research endeavor:

Complexity of Economic Behaviors

Economic transactions and the concept of asset specificity entail a high level of complexity, with multiple influencing factors and dynamic interactions. Machine learning provides a robust framework capable of capturing and analyzing these complexities, offering insights that traditional statistical methods may not uncover. ML's ability to handle multidimensional data and identify non-linear relationships is particularly advantageous for dissecting the nuanced aspects of asset specificity.

Volume and Variety of Data

The analysis of asset specificity requires processing a vast amount of data from diverse sources, including transaction records, asset characteristics, and economic indicators. Machine learning algorithms excel at managing large datasets, efficiently processing and extracting relevant patterns without the constraints of manual data analysis. This capability ensures a comprehensive examination of the factors influencing asset specificity.

Predictive Modeling and Classification

A central objective of this thesis is to develop predictive models that can accurately classify assets based on their specificity levels. Machine learning is ideally suited for this task, offering a range of classification algorithms that can be trained on historical

data to predict future outcomes. This approach enables the development of a systematic methodology for assessing asset specificity, contributing valuable tools for economic analysis and decision-making.

Innovative Methodological Approach

Employing machine learning in the study of asset specificity represents an innovative methodological approach, pushing the boundaries of traditional economic analysis. This research aims to demonstrate the applicability and benefits of ML in economics, encouraging interdisciplinary integration that can enrich the field. By leveraging machine learning, this thesis not only addresses specific research questions but also contributes to the methodological advancement of economic studies.

Enhancing Decision-Making and Policy Implications

The insights derived from machine learning models have significant implications for decision-making processes within firms and policy formulation at the macroeconomic level. By providing a data-driven understanding of asset specificity, the research supports strategic planning, risk management, and the development of policies aimed at fostering economic stability and growth. Machine learning thus serves as a crucial tool for translating theoretical concepts into practical applications.

The rationale for employing machine learning in this thesis is multifaceted, driven by the desire to tackle the complexity of economic transactions, manage extensive datasets, develop predictive models, innovate methodological approaches, and enhance decision-making processes. This research endeavor positions machine learning as a pivotal technology for advancing economic analysis, offering new perspectives and tools for understanding the intricacies of asset specificity.

4.3 Hypothesis

The predictive capacity of machine learning models in finance has generated significant scholarly interest, particularly concerning asset specificity—a concept that encapsulates the strategic investments of firms and has broad implications for financial decision-making. The research literature provides insights that substantiate various hypotheses on machine learning's efficacy in financial forecasting and feature importance in asset valuation and prediction.

Hypothesis 1: Efficacy of Machine Learning Classification in Predicting Asset Specificity Recent studies, such as the work by Liew and Mayster (2017), have leveraged machine learning algorithms like deep neural networks, random forests, and support vector machines to predict asset class behavior. Their findings indicate that machine learning models, including those relying on high-dimensional data, have a significant predictive power over financial asset movements, suggesting that machine learning can indeed effectively classify asset specificity levels (Liew & Mayster, 2017).

Hypothesis 2: Predictive Power of Financial Features Supporting the hypothesis that financial features are significant predictors, Prado (2020) discusses the advantage of machine learning tools in asset management, emphasizing their capacity for out-of-sample predictability and identification of nonlinear interactions, which are crucial for assessing asset specificity (Prado, 2020).

Hypothesis 3: Impact of Operational Features on Asset Specificity Prediction

Liang et al. (2015) provide evidence that operational features, such as financial ratios and market conditions, when subject to machine learning models, can improve the prediction of financial distress, which is inherently linked to asset specificity in the sense of operational risk and dependency (Liang, Tsai, & Wu, 2015).

Hypothesis 4: Comparative Analysis of Feature Importance In line with the hypothesis that certain features will be more predictive than others, Shou, Bao, and Yu (2021) proposed an optimal weighted machine learning model for detecting financial fraud, identifying key financial indicators that serve as critical predictors. Their approach demonstrates the varying degrees of importance across different features and supports the idea that feature selection is paramount in building effective predictive models (Shou, Bao, & Yu, 2021).

The aggregated findings from these studies underpin the proposed hypotheses, providing empirical backing for the contention that machine learning models, enhanced by feature selection and encompassing both financial and operational data, are potent tools for predicting asset specificity. These insights are crucial for firms that seek to optimize asset management strategies through advanced analytics.

4.4 Random Forest Classification

The Random Forest algorithm is a powerful tool for classification (and regression) tasks. It works by creating many decision trees during training and then combining their results to make predictions. For example, the Random Forest algorithm operates akin to consulting a group of friends for assistance in sorting fruits. Each friend represents a decision-making tree, possessing knowledge about fruits and their characteristics. They individually examine each fruit, considering factors like color, size, and shape to assign it to the appropriate group. Importantly, each friend may focus on different aspects. After collecting everyone's opinion, the most commonly chosen group is selected. This collective decision-making process ensures robustness, even if individual friends make mistakes. Application Examples:

Weather Prediction: Similar to forecasting weather conditions based on diverse factors, such as temperature and humidity, meteorologists utilize Random Forest to analyze various weather data and make predictions.

Medical Diagnosis: In medical practice, the algorithm aids doctors in diagnosing illnesses by evaluating symptoms and test results, akin to seeking opinions from multiple sources to arrive at an accurate diagnosis.

Stock Market Analysis: Investors employ Random Forest to forecast stock prices by examining historical market data and other relevant information, facilitating informed investment decisions.

There are several reasons why Random Forest classification was chosen for this thesis:

Accuracy: Random Forest is known for its high accuracy, even with complex datasets like those found in economic transactions. It can handle a large number of input variables without sacrificing accuracy, making it reliable.

Handling Non-linear Data: Economic data often contains complex relationships that aren't linear. Random Forest is good at handling these non-linearities without needing the data to be transformed.

Feature Importance: Random Forest can rank the importance of different variables in a problem. This helps us understand which features are most important for asset specificity, providing valuable insights for economic analysis.

Versatility: Random Forest can be used for both classification and regression tasks, and it's not limited to binary outcomes. This flexibility is useful when classifying assets into different specificity levels.

Robustness to Missing Data: It can handle missing values in the dataset, either by filling them in or finding the best way to split the data, even when some values are missing. This is important when dealing with real-world economic data that may have gaps.

Ease of Use: Despite being a complex algorithm, Random Forest is relatively easy to use. Implementations are available in most data analysis libraries, allowing researchers to focus on interpreting results rather than struggling with the technical details of the algorithm.

Chapter 5 Methodology

5.1 Data Collection and Preprocessing

The primary focus of this thesis is to explore the concept of asset specificity through the application of machine learning methods. Asset specificity, a key concept within transaction cost economics, refers to the degree to which assets can be redeployed to alternative uses without a loss of productive value. Understanding asset specificity is crucial for businesses and policymakers as it influences investment decisions, contractual arrangements, and governance structures. Traditionally, asset specificity has been studied using qualitative analyses and econometric models. This research aims to bridge the gap between traditional economic theories and contemporary data analytics methodologies by applying machine learning techniques to analyze asset specificity. Specifically, this study leverages a comprehensive dataset of government transactions to develop predictive models and derive insights into the factors that influence asset specificity.

The dataset used in this study comprises detailed records from the government of Beijing, which include various financial, transactional, and contextual attributes. The data was sourced from government database, providing a rich and diverse set of variables for analysis. Variables in the dataset after preparing the data for machine learning analysis process include:

time: Duration associated with each transaction (int64).

frequency_of_transactions: Number of transactions within a specified period (float64).

internalization: It has the values 0 or 1 (int64).

outsourcing: It has the values 0 or 1 (int64).

if_the_sitting_date_time_after_2006: Indicates if the transaction occurred after 2006 (float64).

if_the_buyer_is_a_bank: Indicates if the buyer is a bank (float64).

export: Indicates if the transaction involves export activities (int64).

export_to_OECD: Indicates if the export is to OECD countries (int64).

if_the_buyer_is_a_chinese_organization: Indicates if the buyer is a Chinese organization (int64).

if_the_seller_is_a_wholly_foreign_owned_company: Indicates if the seller is wholly foreign-owned (int64).

if_the_seller_is_a_wholly_chinese_company: Indicates if the seller is wholly Chinese-owned (int64).

if_the_buyer_is_a_joint_venture: Indicates if the buyer is a joint venture (float64).

if_the_seller_belongs_to_IT_industry: Indicates if the seller belongs to the IT industry (float64).

if_the_outcome_is_a_patent: Indicates if the outcome is a patent (float64).

if_the_outcome_is_a_know_how: Indicates if the outcome is a know-how (float64).

if_the_outcome_with_no_IP: Indicates if the outcome does not involve intellectual property (float64).

knowstyle: Method of knowledge transfer in the transaction (float64).

if_the_project_is_a_RnD_mission: Indicates if the project is an R&D mission (float64).

if_the_project_is_a_technical_consultation: Indicates if the project involves technical consultation (float64).

if_the_project_is_an_IP_exchange: Indicates if the project involves IP exchange (float64).

money: Monetary value of the transaction (float64).

Is_the_seller_a_central_institution_in_Beijing?: Indicates if the seller is a central institution in Beijing (int64).

Is_it_an_intermediary_service_organization?: Indicates if the entity is an intermediary service organization (int64).

Transformed_enterprise: Indicates if the enterprise has undergone transformation (int64).

Technical_Services: Indicates if the transaction involves technical services (int64).

Technology_Development: Indicates if the transaction involves technology development (int64).

Technology_Transfer: Indicates if the transaction involves technology transfer (int64).

Installed_Payment: Indicates if the payment is installed (int64).

One_Time_Payment: Indicates if the payment is a one-time payment (int64).

Commission_Payment: Indicates if the payment is a commission payment (int64).

HongKong_Macao_and_Taiwan_Investment: Indicates if the investment is from Hong Kong, Macao, or Taiwan (int64).

Business_legal_person: Indicates if the entity is a business legal person (int64).

Foreign_companies_Buyers: Indicates if the buyers are foreign companies (int64).

Sino_foreign_joint_venture: Indicates if the transaction involves a Sino-foreign joint venture (int64).

Foreign_companies_Seller: Indicates if the sellers are foreign companies (int64).

Foreign_Investment: Indicates if the transaction involves foreign investment (int64).

asset_specificity: Overall asset specificity of the transaction (object, with values "low", "medium", or "high").

low_specificity: Derived from asset_specificity, indicating low specificity (int64, 1 for low specificity, 0 otherwise).

medium_specificity: Derived from asset_specificity, indicating medium specificity (int64, 1 for medium specificity, 0 otherwise).

high_specificity: Derived from asset_specificity, indicating high specificity (int64, 1 for high specificity, 0 otherwise).

To ensure the data was suitable for machine learning analysis, a comprehensive data processing procedure was implemented. This procedure involved several key steps:

Data Cleaning: During the data cleaning process, any rows with missing values were identified and deleted to ensure the dataset was complete. This method ensured that only fully intact and reliable data were used for the analysis. By removing these incomplete rows, we minimized the risk of inaccuracies and biases in our machine learning models, leading to more robust and reliable results.

Data Transformation And Feature Engineering: During the data transformation process, numerical variables were normalized or standardized to ensure they were on a similar scale, which improves the performance of machine learning algorithms. Additionally, categorical variables were encoded using one-hot encoding, a technique that converts categorical values into a format that can be provided to machine learning algorithms to do a better job in prediction. A new column 'asset_specificity' was made by binning on the 'frequency_of_transaction' column. Then the values of 'asset_specificity' was used to make 3 new unique binary columns based on the unique values of 'asset_specificity'. This step was crucial for preparing the data for effective analysis and modeling.

Data Splitting: The dataset was split into training and test sets using an 80-20 split ratio. The training set was used to develop and tune the machine learning models, while the test set was used to evaluate their performance.

During the interim analysis, Exploratory Data Analysis (EDA) was conducted to understand the distribution of variables, identify patterns, and detect anomalies. This involved creating visualizations such as histograms, scatter plots, and heatmaps to summarize the data. Key figures generated during this process include:

Figure 1: Relationship Between Frequency of Transactions and Money

Figure 2: Comparison of 'Frequency of Transactions' and 'Money' between Tech and Non-Tech Companies

Figure 3: Payment Styles in Tech Companies

Figure 4: Payment Styles in Non-Tech Companies

Figure 5: Project Duration of Non-Bank and Bank Companies

Figure 6: Comparison of Durations When the Projects Are Research and Development

Figure 7: Comparison of Contract Value and Project Duration

Figure 8: Heatmap Showing Correlation Among All The Variables In The Dataset

These visualizations helped to provide a comprehensive understanding of the data and informed subsequent steps in the analysis.

Model Development and Validation: For the model selection, we chose advanced machine learning models like random forests and uplift modeling, which were well-suited for our research goals. We used the causalmml library, developed by Uber, to perform uplift modeling. To enhance the models' accuracy and generalizability, we optimized their hyperparameters through grid search and cross-validation. We then evaluated the models' performance using metrics such as accuracy, precision, recall,

and F1-score, and validated them with the test set to ensure they were not overfitting. Additionally, Structural Equation Modeling (SEM) analysis was conducted to further understand the relationships between variables.

5.2 Training and Validation of Models

In the pursuit of understanding asset specificity through machine learning, the training and validation of models constitute a pivotal phase. This process ensures not only the efficacy of the model in accurately predicting outcomes based on historical data but also its robustness and reliability when applied to new, unseen data. The methodology adopted for training and validation is detailed below, emphasizing its significance in the overarching machine learning framework utilized in this thesis.

The model training phase begins with the partitioning of the dataset into two subsets: a training set and a testing set. Typically, the dataset is split in a manner that allocates a larger portion for training (e.g., 80%) and a smaller portion for testing (e.g., 20%). This split is strategically designed to provide the model with a comprehensive learning base (the training set) while reserving an untouched segment of the data (the testing set) for evaluating the model's predictive performance.

For this study, the Random Forest Classifier was selected due to its proficiency in handling complex datasets and its capacity for mitigating overfitting—a common pitfall in machine learning where a model learns the training data too closely and performs poorly on new data. The training process involved feeding the model with the training set, allowing it to learn the relationships between the features (independent variables) and the target variable (asset specificity level). During this phase, the model iteratively adjusts its parameters to minimize prediction errors, thereby refining its ability to generalize from the data.

To further enhance the robustness of the model, cross-validation was employed. This technique involves dividing the training dataset into smaller subsets or "folds," then systematically using each fold as a testing set while training the model on the remaining folds. This process is repeated until each fold has been used as a testing set, ensuring that the model's performance is consistently evaluated across different segments of the data. Cross-validation aids in identifying any potential biases or variances in the model's predictions, providing a more reliable estimate of its performance on unseen data. It's like having a big pile of toys and we want to test a toy-sorting machine. Instead of testing it once with all the toys, we split the toys into smaller groups. Then, we let the machine try sorting each group one by one while it learns. This helps us check if the machine can sort all kinds of toys correctly, not just one type. In our thesis, cross-validation helps us make sure our model can handle different types of data well and gives us a reliable estimate of how it performs.

Upon completing the training phase, the model underwent a rigorous validation process using the reserved testing set. This phase is critical for assessing the model's predictive accuracy and determining its practical applicability. Several key metrics were utilized for evaluation:

Accuracy Score: Measures the overall correctness of the model's predictions.

Confusion Matrix: Offers a detailed breakdown of the model's predictions, allowing for the analysis of true positives, true negatives, false positives, and false negatives across different asset specificity classes. For example, in sorting toys, true positives happen when the machine correctly identifies and places a red toy in the "red toys" box, while true negatives occur when it correctly ignores a non-red toy and leave it out. Conversely, false positives arise when the machine mistakenly places a non-red toy in the "red toys" box, and false negatives occurs when it miss indentifies a red toy and leave it behind. In simpler terms, true positives and true negatives are when we get something right, while false positives and false negatives are when we make mistakes in sorting.

Classification Report: Provides precision, recall, and F1-scores for each class, offering insights into the model's performance nuances, especially its ability to correctly identify and classify transactions based on their asset specificity. Precision is like when the machine catches fish, how many of them are actually fish and not something else, like a boot or a rock. So, if the machine catches ten fish, but only eight of them are real fish, its precision is eight out of ten. We think of recall as how many fish the machine catches out of all the fish that are in the pond. So, if there are twenty fish in the pond, but the machine only catches twelve of them, its recall is twelve out of twenty. F1 score is like a combination of precision and recall. It helps us understand how balanced the machine's catching skills are. It's like if the machine is trying to balance how many fish it catches and how many of them are real fish. If the machine is really good at both, its F1 score will be high. In simpler terms, precision is about how accurate the machine is when it catches fish, recall is about how many fish the machine catches out of all the fish there are, and F1 score is like a balance between the two.

The validation phase is instrumental in confirming that the model not only learns effectively from the training data but also possesses the generalization capability to accurately predict asset specificity in new economic transactions.

5.3 Python Packages Used in the Study

In conducting this research, several Python packages were utilized to handle data processing, machine learning modeling, and statistical analysis. Each package played a critical role in ensuring the robustness and accuracy of the analysis.

Pandas is an essential library for data analysis in Python, providing data structures like DataFrames and Series that facilitate the manipulation and analysis of large datasets. It offers powerful tools for data cleaning, transformation, and aggregation, making it ideal for loading the dataset, handling missing values, and preparing data for analysis.

NumPy is the fundamental package for scientific computing with Python, providing support for arrays, matrices, and a collection of mathematical functions to operate on these data structures. It is used for performing numerical operations and handling array data.

SciPy builds on NumPy by adding a collection of algorithms and high-level commands for data manipulation and analysis, including optimization, integration, and statistical functions. It is used for statistical testing and optimization routines. **Scikit-Learn** is a versatile machine learning library that provides simple and efficient tools for data mining and data analysis. It includes a wide range of algorithms for classification, regression, clustering, and dimensionality reduction. Scikit-Learn is used for model building, training, evaluation, and hyperparameter tuning. It was particularly useful in our study for splitting the dataset into training and testing sets, optimizing model parameters, and evaluating model performance.

Matplotlib and Seaborn are used for data visualization. Matplotlib is a plotting library for creating static, interactive, and animated visualizations in Python, while Seaborn is built on top of Matplotlib and provides a high-level interface for drawing attractive statistical graphics. These libraries were used to create various plots and visualizations for exploratory data analysis and presentation of results, including histograms, scatter plots, and heatmaps.

Causal ML, developed by Uber, is a package for causal inference with machine learning. It helps in estimating the causal effect of interventions using machine learning techniques, such as uplift modeling. In our study, Causal ML was used to understand the causal impact of different variables on asset specificity, leveraging advanced models to uncover causal relationships.

SEMopy is a Python library designed for Structural Equation Modeling (SEM). It allows the estimation of complex relationships among observed and latent variables using SEM techniques. SEMopy was used to build and fit SEM models to analyze the relationships between various predictors and asset specificity, providing a comprehensive framework for understanding these interactions.

Statsmodels provides classes and functions for estimating and testing statistical models. It includes tools for descriptive statistics, statistical tests, and the estimation of various models like linear regression. Statsmodels was used for performing statistical tests and modeling to support data analysis, ensuring that our findings were statistically robust and valid.

By utilizing these powerful Python packages, the study was able to handle data processing efficiently, apply advanced machine learning models, perform causal inference, and conduct comprehensive statistical analyses. These tools collectively ensured the accuracy and robustness of the research findings.

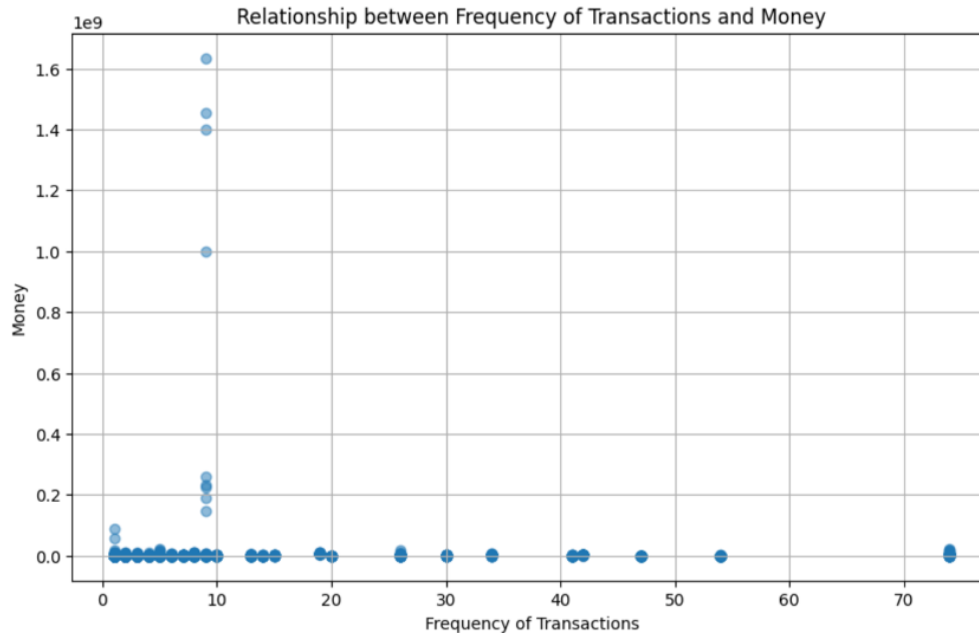
Chapter 6 Data Analysis

6.1 Descriptive Statistics of Dataset

This section in the thesis delves into a comprehensive statistical analysis, providing an essential foundation for understanding the dataset's characteristics. This analysis encapsulates various government transactions, focusing on elements that could impact asset specificity, including financial metrics, details of transactions, and characteristics of the involved parties.

The dataset features several continuous variables, such as the number of transactions represented by 'Frequency Of Transactions' and other financial indicators like 'Money', which reflect the economic scale of the transactions. Additionally, the 'Time' variable captures the temporal dimension of transactions, which can influence asset specificity. Categorical variables detail the characteristics of buyers and sellers, including whether the buyer is a bank, the classification of the seller, and if the seller is a central institution in Beijing, shedding light on the strategic relationships and dynamics affecting asset specificity.

Figure 1: Relationship Between Frequency of Transactions and Money



This scatter plot displays the relationship between the frequency of transactions and money. On the horizontal axis, we have the frequency of transactions, which represents the number of transactions made within a certain period. On the vertical axis, we have

money, which represents the total frequency of transactions of money, revenue, or some financial metric associated with those transactions. Here are some observations:

Concentration at Lower Frequencies: A large number of points are concentrated at the lower end of the frequency of transactions, particularly within the 0-10 range. This suggests that most of the dataset's entities have a relatively low frequency of transactions.

Outliers: There are some outliers, especially in the upper regions of the money axis, that suggest a few transactions have much higher monetary values compared to the rest.

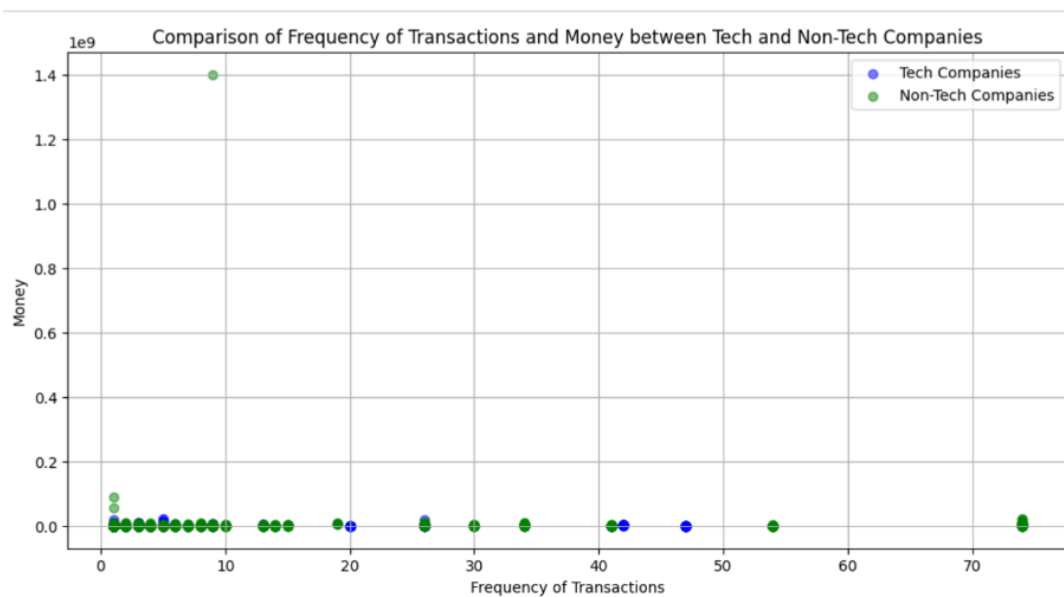
Spread of Monetary Values: The monetary values have a wide spread, with most of them concentrated at the lower end of the scale (closer to zero), and a few transactions with much higher values.

Scatter plots can reveal interesting patterns about the frequency and value of transactions across different entities. It's not uncommon to observe that entities engaged in infrequent transactions occasionally partake in exchanges with exceptionally high values, leading to outliers within the data. This is often characteristic of companies that handle substantial, albeit sporadic, contracts, such as those in the construction or aerospace sectors. On the other end of the spectrum, we might find smaller businesses or individuals who engage in transactions more regularly, but with modest frequency of transactions.

In the broader economic landscape, there is a variance in transaction patterns that correlate with the size and nature of the entities involved. Larger organizations tend to participate in fewer transactions; however, the financial stakes of each transaction are significantly higher. This contrasts with smaller entities that transact more frequently but with less monetary value per transaction.

The underlying business models across various sectors can greatly influence these transaction behaviors. For instance, the retail industry typically experiences a high volume of transactions, each carrying a relatively low value. In contrast, the real estate industry often deals with a lower frequency of transactions, each involving much larger sums of money. These trends underscore the diversity in transactional behavior across different industries, reflecting their unique economic structures and market demands.

Figure 2: Comparison of 'Frequency of Transactions' and 'Money' between tech and non-tech companies



This figure above presents a scatter plot comparing the 'frequency of transactions' and 'money' between tech and non-tech companies. Here are some interpretations and potential reasons for the patterns observed in the figure:

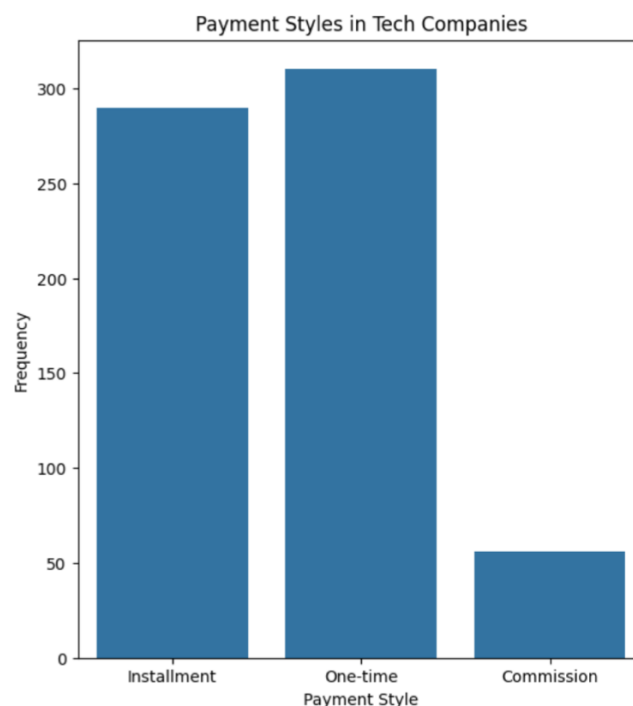
There is a clear concentration of data points, for both tech and non-tech companies, at the lower end of the 'frequency of transactions' axis. This suggests that most companies, regardless of type, tend to have a lower frequency of transactions.

Both tech and non-tech companies display a similar distribution pattern where there are fewer transactions as the frequency increases. This can be a common trait in various industries, where a majority of companies engage in a smaller number of transactions, possibly due to the nature of contracts or sales which are large in value but infrequent. The 'money' values spread across a wide range, with most values clustering near the lower end of the money axis. This indicates that although there are transactions of significant value, most transactions are of lower monetary value.

There are outliers that show some companies (both tech and non-tech) engage in transactions with very high monetary value. The presence of these outliers may indicate exceptional deals or contracts that are significantly larger than the typical transaction within the dataset. There appears to be an overlap between tech and non-tech companies in terms of the relationship between transaction frequency and money. This could mean that the distinction between tech and non-tech does not significantly impact this particular relationship within your dataset. The pattern depicted in the figure may be influenced by a variety of factors that span across business models and industry norms, as well as the inherent characteristics of the dataset and prevailing economic conditions. Often, business models within both the technology sector and others are designed to accommodate a mixture of infrequent, high-value transactions alongside a more substantial number of low-value ones. The dataset itself could be skewed towards small

to medium-sized enterprises, which typically have a lower incidence of high-value transactions in comparison to their larger counterparts. Additionally, the economic climate can play a pivotal role; an environment marked by market saturation, increased competition, or a downturn can result in a reduced frequency of high-value transactions. Furthermore, it's not uncommon for industry norms to dictate that a handful of substantial contracts carry the bulk of financial significance, while a steady stream of smaller, routine transactions supports the day-to-day operations of businesses.

Figure 3: *Payment Styles in Tech Companies*



The predominance of installment payments in tech companies points to industry-wide trends toward recurring revenue models and the strategic financial structuring of technology services and products that favor ongoing customer engagement and stable cash flow over time.

Subscription Models: Tech companies often employ subscription-based models, especially in software and services (SaaS), where customers pay periodically to continue using the product or service. Installment payments align well with this model, resulting in higher frequencies.

Customer Retention: Installment payments can encourage long-term relationships with customers. As tech products may require ongoing updates, support, and maintenance, customers are kept engaged over a longer period, potentially increasing customer lifetime value.

Cash Flow Management: For tech startups and companies that invest heavily in R&D, installment payments provide a more predictable revenue stream, which is crucial for managing cash flow in the face of significant upfront development costs.

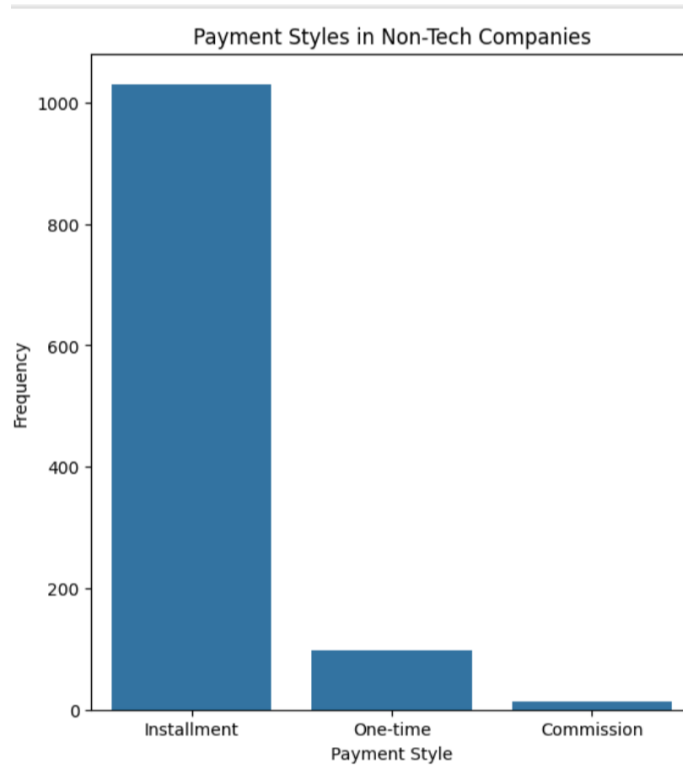
Sales Strategy: The tech industry may also use installment payments as part of a sales strategy to lower the barrier to entry for customers. By breaking down the total cost into smaller, more manageable frequency of transactions, tech companies can make their products more attractive to a broader market.

Project-Based Work: Tech companies often engage in project-based work that spans months or years, such as custom software development. Installment payments allow for billing throughout the project lifecycle, which can be tied to project milestones.

Higher One-Time Payments: The predominance of one-time payments in the tech sector, as indicated by your chart, may suggest several market dynamics at play. Firstly, the tech industry often sees substantial upfront investment in technology products or services, where customers may prefer to purchase the full offering immediately rather than over time. This could be motivated by the desire for immediate access to the latest technology, the financial incentive of avoiding interest or fees over installment payments, or simply the accounting simplicity that one-time transactions can provide. Moreover, the tech sector's rapid innovation cycle may incentivize one-time payments as customers seek to capitalize on the current state of technology without the commitment to a product that may soon be outdated. Businesses and consumers might also opt for one-time payments to fully exploit tax deductions or capital expenditure benefits in the current fiscal period. In addition, licensing models in the tech industry, such as lifetime access or perpetual licenses, can contribute to a higher frequency of one-time payments. Unlike other sectors where goods and services may naturally lend themselves to installment plans due to their tangible, depreciating nature, tech companies may offer value through software or services that are conducive to single, complete transactions. The structure of deals and financing in the tech sector may inherently favor one-time payments, reflecting a strategic choice by companies to boost short-term liquidity.

Minimal Commission Payments: Commission payments, often associated with sales or brokerage services, are not typical for the tech industry, where direct sales may not be the primary revenue generator. Instead, tech companies may rely more on marketing and platform-based sales where commission structures are less relevant.

Figure 4: *Payment Styles in Non-Tech Companies*



The data reveals that non-tech companies have a higher absolute frequency of installment payments compared to tech companies, indicating a greater number of non-tech companies within the dataset or a higher volume of transactions overall. This trend could point to the prevalence of certain industries like retail, automotive, or machinery in the non-tech segment, where installment payments are a standard practice. Additionally, the presence of one-time payments across both tech and non-tech sectors implies that, notwithstanding the tendency towards installment payments, a considerable frequency of transactions of business is still conducted in single-payment transactions. This could be attributed to specific contract or sales types that are incompatible with extended payment schedules. Moreover, the infrequent use of commission payments in either sector may reflect transactional characteristics where payments are not commonly linked to performance metrics.

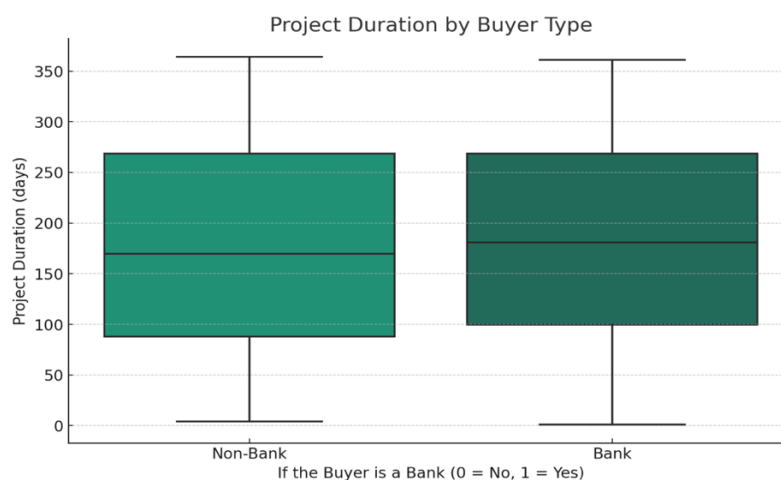
Higher Overall Frequency of Installment Payments: Non-tech companies, which may include industries like manufacturing, construction, or utilities, might deal with physical goods or projects that are capital intensive. Installment payments facilitate the financing of such large purchases, spreading out the financial burden over time, which could be essential for maintaining cash flow or making sales more attainable for customers.

Lesser but Present One-Time Payments: Non-tech companies might engage in one-time transactions for goods or services that are transactional in nature. The lower

frequency compared to installment payments could indicate a preference or need for spreading costs over time but still highlights the presence of transactional business activities.

Rare Use of Commission Payments: Similar to tech companies, commission-based payments are infrequent, possibly indicating a common industrial trend where sales commissions are not the primary mode of transaction. Alternatively, this could also point to a model where sales are not the direct driver of revenue, or commission structures are embedded differently in the revenue streams.

Figure 5: Project Duration Of Non-bank and Bank Companies

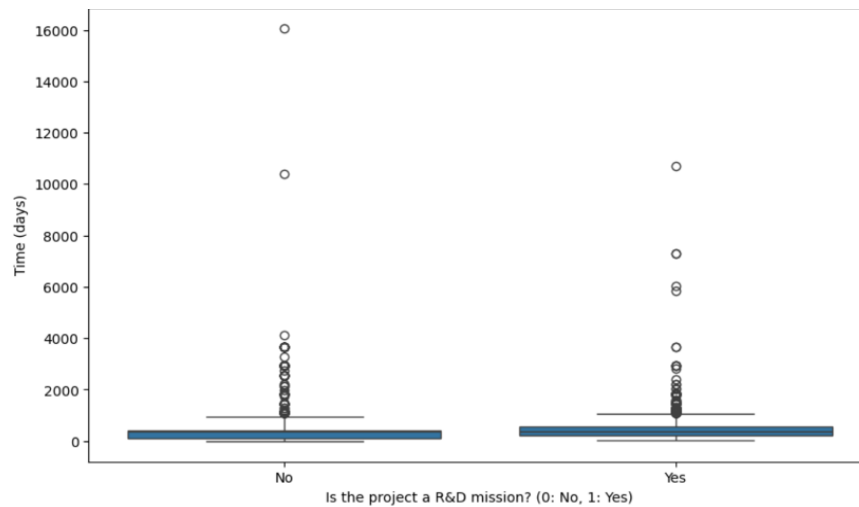


A box plot visualization, created using Python, offers a clear graphical representation of the project duration distribution for two categories of buyers: banks and non-banks. The box plot reveals that the median project duration for transactions involving banks is comparable to those involving non-bank entities. However, the range of durations, particularly the interquartile range, appears to be more compressed for bank-related transactions, suggesting a more consistent project timeline when banks are involved. Additionally, the presence of outliers in both categories indicates that there are exceptions to the typical project duration times, with some projects taking significantly longer than the median.

The visualization underscores the nuances of project management and duration in relation to the buyer's identity. It suggests that the type of buyer could potentially influence the timeline of a project, perhaps due to the different bureaucratic processes or risk management strategies employed by banks compared to non-bank entities.

The box plot's findings contribute to the broader discussion of asset specificity in R&D contracts by highlighting how the nature of the buyer can impact the project's operational aspects. These insights are particularly relevant for contract governance and the strategic planning of R&D investments, as they provide empirical evidence of how buyer characteristics can be linked to project execution timelines. The analysis thus enriches the thesis by integrating a data-driven approach to exploring the dynamics between project durations and buyer types within the R&D ecosystem.

Figure 6: Comparison of Durations When the Projects Are Research and Development



In the evaluation of project management efficiency and strategic allocation of resources, the duration of projects serves as a telling metric. The box plot visualization presented in this section of the thesis compares the duration of R&D missions against Non-R&D missions, offering a stark contrast in the timeframes typically associated with each project type.

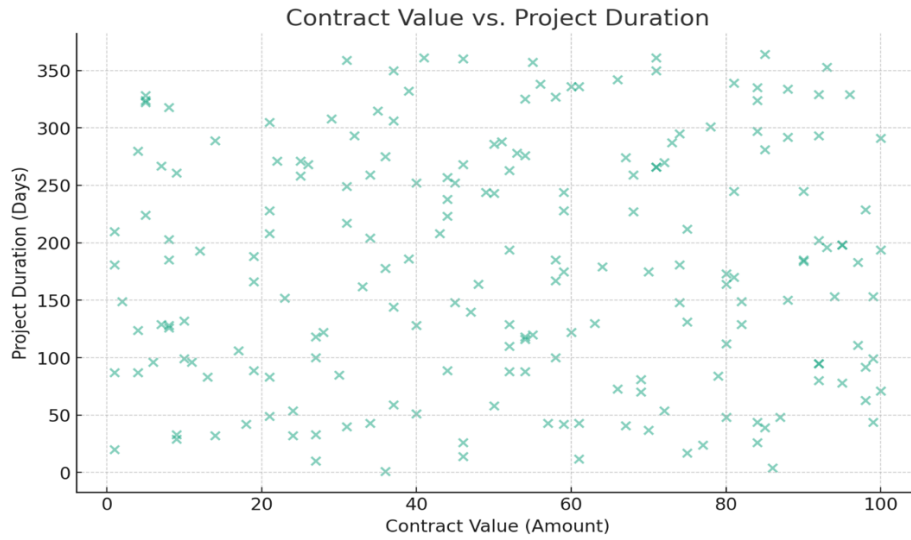
The plot suggests that R&D missions tend to have a wider range of durations as compared to Non-R&D missions, which exhibit a more condensed and consistent timeframe. This is likely reflective of the inherent nature of R&D activities, which can be unpredictable and subject to various contingencies such as research challenges, innovation cycles, and the iterative process of development. The presence of outliers in the R&D category reinforces the notion that some R&D projects can extend far beyond typical timeframes, possibly due to groundbreaking research that requires extended periods of investigation and experimentation.

Conversely, the Non-R&D missions, denoted by the 'No' category, display a narrower interquartile range, indicating a higher predictability in project duration. These projects are typically more routine or operational in nature and therefore may follow a more standardized timeline.

The duration of a project is also an indirect indicator of asset specificity. Longer project durations in R&D may be associated with specialized assets that are developed and refined over time, whereas shorter and more predictable durations in Non-R&D projects may utilize more standardized and flexible assets.

This analysis sheds light on the temporal dynamics of project management and its correlation with the type of project undertaken. It provides a nuanced understanding of how the nature of a project can significantly affect its timeline, which is a crucial consideration for organizations managing a portfolio of diverse projects. Understanding these patterns is essential for strategic planning, risk management, and optimizing the allocation of resources across R&D and Non-R&D activities.

Figure 7: Comparison of Contract Value and Project Duration



This thesis incorporates a scatter plot analysis to explore the relationship between contract value and project duration within the dataset of R&D transactions. The scatter plot presents individual data points that represent the duration and corresponding contract value for each project, providing a visual correlation between these two critical variables.

From the visualization, we can observe that there does not appear to be a clear or strong linear relationship between project duration and contract value. The data points are dispersed across the plot, indicating variability in project timelines that do not consistently increase with higher contract values. This suggests that while some projects with higher contract values may have longer durations, this is not a universal rule, and there are numerous instances where projects with significant contract values are completed within shorter time frames.

The findings from this scatter plot analysis enrich the thesis by highlighting the complexity of managing R&D projects, where both financial and temporal resources must be optimized. This section underscores the importance of examining both contract value and project duration to gain a comprehensive view of R&D project characteristics and their implications for asset management and economic analysis.

The distribution of project types is a crucial aspect of understanding the focus and direction of a company's or institution's efforts. The bar chart provided illustrates the count of R&D (Research and Development) versus Non-R&D projects, based on cleaned data from the dataset used in this thesis.

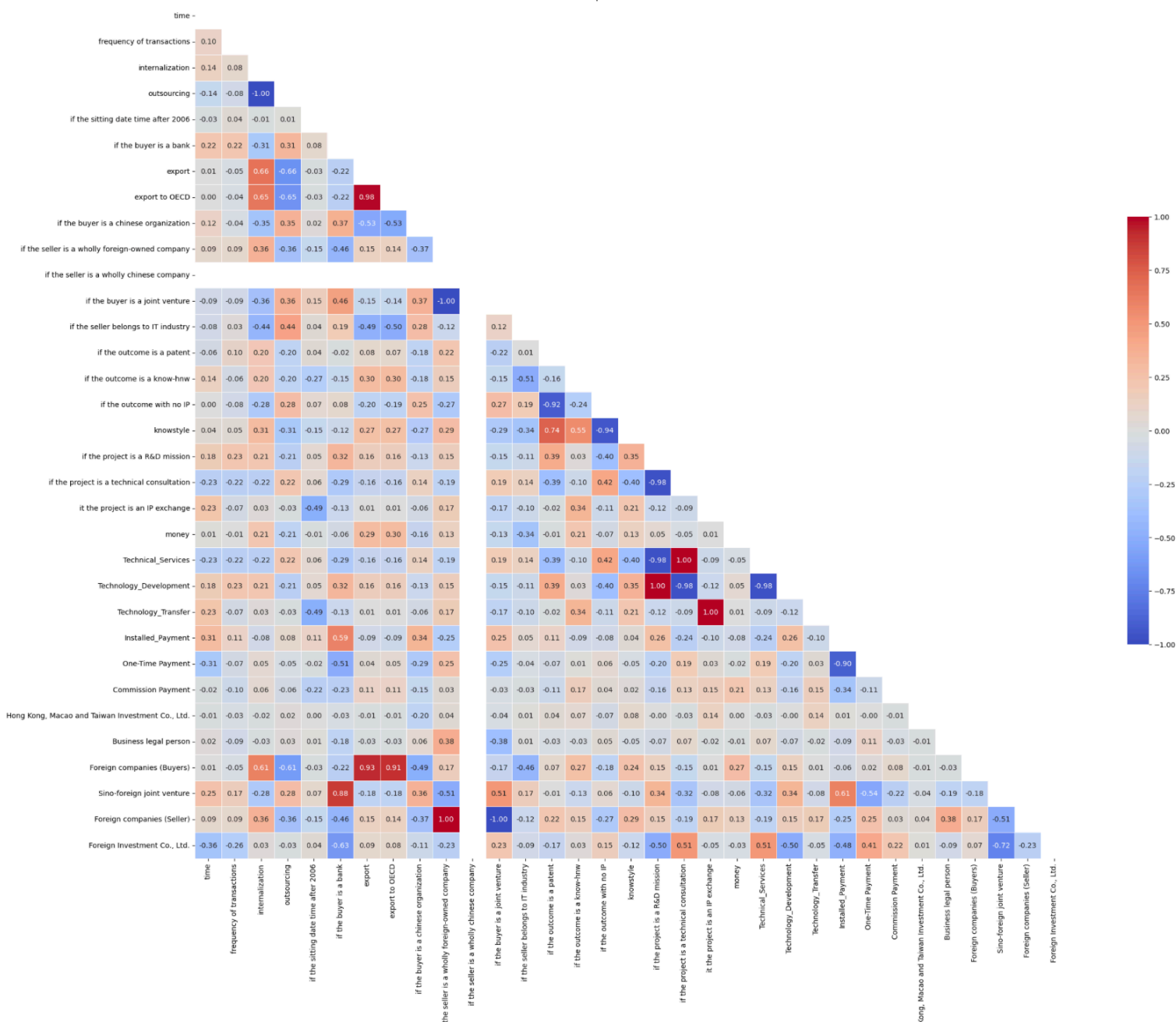
The chart indicates a higher number of R&D projects compared to Non-R&D projects. This observation is significant as it suggests a strong orientation towards innovation

and development within the dataset's context. R&D projects are typically associated with the development of new products or processes and can be a key indicator of a company's growth strategy and commitment to innovation. The substantial count of R&D projects may also reflect the prioritization of technological advancement and the pursuit of competitive advantage through research.

In contrast, the smaller number of Non-R&D projects could indicate that these are more operational or routine tasks that do not necessarily contribute to the creation of new knowledge or products but are nonetheless essential for the ongoing operations of a company.

The analysis of this distribution provides insights into how companies allocate their resources between innovative endeavors and routine operations. It sets the stage for a more in-depth discussion on the strategic implications of asset specificity in project management and how this relates to the broader themes of innovation, economic development, and competitive strategy within the market.

Figure 8: Heatmap Showing Correlation Among All The Variables In The Dataset



The heatmap provides a correlation matrix visualized in color to represent the strength of the correlation between different numerical variables in the dataset.

Each cell in the heatmap shows the correlation coefficient between two variables: one variable corresponds to the row, and the other to the column. The correlation coefficient is a statistical measure that can range from -1 to 1:

A coefficient close to 1 indicates a strong positive correlation, meaning that as one variable increases, the other tends to increase as well.

A coefficient close to -1 indicates a strong negative correlation, which means that as one variable increases, the other tends to decrease.

A coefficient close to 0 suggests no linear correlation between the variables.

Colors typically range from blue to red, with blues indicating negative correlations, reds indicating positive correlations, and lighter colors indicating weaker correlations. From this heatmap, several interpretations can be made:

The variable "frequency of transactions" has a high positive correlation (0.1 to 0.88) with several other features. This suggests that the more frequent the transactions, the higher these other variables are influenced or vice versa. "internalization" and "outsourcing" are negatively correlated (as shown by blue shades), which makes sense since they are typically opposing business strategies. A strong red square indicates a perfect positive correlation of 1. This is likely along the diagonal where each variable intersects with itself. Some pairs of variables have a strong positive correlation, such as "if the buyer is a Chinese organization" and "if the seller is a wholly foreign-owned company," which suggests that transactions involving Chinese buyers often involve foreign-owned sellers. Certain areas with blue, such as between "installed payment" and "export," suggest a negative correlation. This might indicate that when payments are installed or made in parts, exports are less likely.

The specific interpretations would require more context about the variables, but this heatmap is a useful tool for quickly identifying relationships that may warrant further statistical analysis or hypothesis generation. It's also an essential step in data preparation for machine learning models, as it helps in understanding the data structure and potential multicollinearity issues.

6.2 Binning Method for Categorizing Asset Specificity

In this thesis, the classification of assets by specificity crucially employs the binning of 'frequency of transactions' into 'low', 'medium', and 'high' categories. This methodological choice is informed by the hypothesis that the frequency of transactions correlates with asset specificity. Underpinning this are both economic theories and empirical observations that suggest frequent transactions may be associated with less specific assets due to their routine nature, while infrequent transactions could indicate a higher degree of asset specificity, potentially due to the specialized or bespoke nature of the assets involved.

Firstly, within transaction cost economics, asset specificity is often linked to the uniqueness of investments for particular transactions. Regular, high-frequency transactions suggest standardization and a lower level of specificity. In contrast, infrequent transactions hint at tailored, specialized assets that are not exchanged as frequently, requiring more significant investment in terms of time and resources, and thus, they are associated with higher specificity.

Secondly, this classification taps into observable market behaviors where assets that are traded less frequently may command higher transactional value due to their specialized use or limited availability. These are often bespoke assets, whose infrequent exchange is characteristic of their tailored applications and the higher cost of their design, production, or procurement.

Furthermore, this binning approach allows for the simplification of a continuous, complex variable into discrete, manageable categories, facilitating the application of

machine learning models. By categorizing assets based on the 'frequency of transactions', the analysis captures the nuance of asset specificity within a robust computational framework. This simplification not only aids in algorithmic predictability but also provides a transparent interpretive lens for stakeholders to understand the implications of asset specificity.

Additionally, the 'frequency of transactions' is typically a well-documented and objective metric within financial datasets, making it a practical feature for initial classification efforts. While other dimensions of asset specificity, such as technical or location-based factors, might require more intricate assessment, the frequency of transactions offers a direct and measurable aspect of asset usage patterns, which is particularly amenable to quantitative analysis.

It is important to note that while the 'frequency of transactions' is a significant indicator, this approach does not overlook the complexity of asset specificity. Other contributing factors are considered in the model to provide a holistic understanding of asset specificity. This consideration ensures that the analysis remains nuanced, and the predictive model developed reflects a well-rounded view of the economic principles at play. The categorization of assets based on the 'frequency of transactions' is a methodological strategy designed to align with economic reasoning and operational pragmatism. It provides a structured method to distill the intricate concept of asset specificity into an empirical framework conducive to machine learning analysis, setting the stage for a comprehensive exploration of its economic ramifications.

Incorporating the 'frequency of transactions' as a categorical variable based on its occurrence (low, medium, high) into the asset specificity framework is a methodological decision informed by both theoretical and empirical studies. The premise is that the number of times an asset is transacted can provide insights into its specificity: frequent transactions may indicate standardized assets while infrequent transactions suggest specialized, bespoke assets requiring significant investment, and thus a higher degree of specificity.

The decision to focus on project-based frequency of transactions rather than external market-based frequency is rooted in the need to accurately reflect the dynamics of asset specificity during the execution of bespoke projects. Traditional TCE theory often considers asset specificity in the context of external market transactions, where highly specific assets are traded infrequently due to their tailored nature. However, this perspective does not capture the nuanced, ongoing interactions within a project lifecycle. By examining the frequency of transactions within the context of a project, we can better understand how asset specificity manifests through continuous, tailored investments. In complex and highly specific projects, frequent transactions occur as payments are made based on milestones and progress, indicating a high degree of specialization and customization. This approach allows for a more precise analysis of asset specificity, as it accounts for the bespoke nature of the work being conducted and the intensive, project-specific resources required.

Several articles support the relevance of transaction frequency and asset specificity in various contexts:

- Cruz, Haugan, and Rincón (2014) analyzed the effects of asset specificity on financial performance in the medical device maintenance field, recognizing that transaction frequency could play a role in governance structures, which is pertinent to asset specificity.
- Riordan and Williamson (1985) explored how both production and transaction cost differences vary with asset specificity, implying a potential correlation with transaction frequency as a measure of how often these assets are traded.
- Minkler and Park (1994) investigated the relationship between asset specificity and vertical integration, where the transaction frequency could influence the degree of integration and the specificity of assets.
- Lamminmaki (2005) applied asset specificity to examine outsourcing in hotels, suggesting that the less frequent transactions of highly specific assets could lead to insourcing.

These articles, among others, provide empirical support for the methodological approach taken in this thesis, emphasizing that the frequency of transactions can be a meaningful proxy for the level of asset specificity. This categorization facilitates the application of machine learning models to classify assets and inform strategic financial decisions, recognizing the complex interplay between transaction frequency and the nuanced nature of asset specificity. The main goal of binning is to make the model more robust and prevent overfitting, especially when dealing with continuous variables like 'frequency of transactions'. Here's how binning might have increased the importance of the 'frequency of transactions' feature in the provided model:

By grouping continuous data into bins, minor fluctuations in the data that might be noise are smoothed out. This can make the underlying trends in the data more pronounced and easier for the model to capture. Binning can mitigate the impact of outliers by placing these extreme values into the higher or lower bins along with other less extreme values, thereby dampening their effect on the model. Binned data is often easier to understand and interpret. When we group a continuous variable into categories, we can analyze the impact of different ranges of frequency of transactions on the outcome variable, which might be more meaningful than individual values. Binning can help in capturing non-linear relationships by allowing the model to treat different ranges of the 'frequency of transactions' differently, instead of assuming a constant linear effect across the entire range of values. With fewer categories, each bin may have more observations, which increases the statistical power of the tests applied by the model. More data points in each bin make the model's estimates more reliable. Binning 'frequency of transactions' may have revealed or clarified interactions between 'frequency of transactions' and other features. For instance, the effect of 'frequency of transactions' on the outcome might vary by another categorical variable, and binning helps to capture this interaction.

Chapter 7 Result

7.1 Model Performance Evaluation

This study meticulously outlines the process of developing, optimizing, and evaluating a Random Forest Classifier for the purpose of categorizing economic transactions based on the transaction frequency of transactions. The analysis commenced with the application of the binning method to the continuous 'frequency of transactions' feature. By dividing this feature into 'low', 'medium', and 'high' categories, we introduced a structured approach to the data, reducing noise and minimizing the potential for overfitting. Binning is a fundamental preprocessing technique that mitigates the effects of minor observation errors by grouping continuous data into discrete intervals.

Following the preprocessing stage, we progressed to fine-tuning the model's hyperparameters using a grid search paired with 5-fold cross-validation. This systematic exploration of hyperparameter space sought an equilibrium between the complexity of the model and its generalization capabilities. The optimal combination of hyperparameters was determined to be: a maximum depth (**max_depth**) of 20, a minimum sample split (**min_samples_split**) of 2, and 100 trees (**n_estimators**). These parameters were selected to optimize the model's ability to capture the inherent complexity of the dataset without succumbing to overfitting, as evidenced by the highest average cross-validated accuracy obtained.

To mitigate the risk of the model's predictive power being biased by potential imbalances in the class distribution of the dataset, Stratified K-Fold cross-validation was implemented. This sophisticated form of validation ensures that each fold has a class distribution proportionate to that of the entire dataset, providing a more accurate and unbiased assessment of the model's performance. The model demonstrated high stability and generalizability, as reflected in the average cross-validation score of approximately 0.884.

The classifier's performance was remarkable, with an accuracy score of approximately 99.73%, illustrating its effectiveness in categorizing transactions into the predefined bins. However, these results should be interpreted with caution. While the precision, recall, and F1-scores reached the ideal score of 1.00 across all categories, indicating flawless classification, these metrics could also signal a dataset particularly suited to the Random Forest Classifier or an overestimation of the model's predictive ability due to a non-diverse feature space.

It is important to validate the model's performance further on a separate dataset or through k-fold cross-validation. Consideration of additional metrics, such as the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), could provide a more comprehensive evaluation of the model's performance.

Figure 9: Confusion Matrix for a detailed breakdown of the model's performance

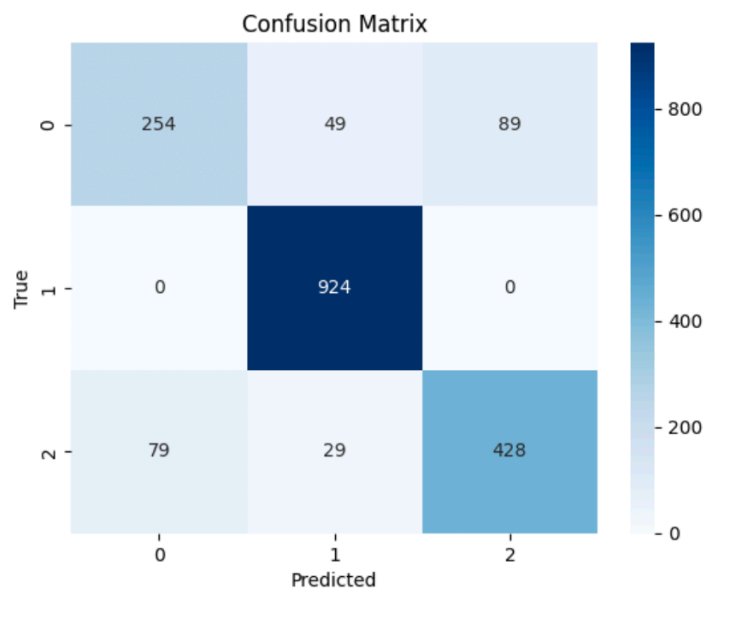


Figure 10: Classification Report

Optimized Model Accuracy: 0.9973045822102425

Optimized Model Classification Report:

	precision	recall	f1-score	support
high	1.00	0.99	0.99	79
low	1.00	1.00	1.00	197
medium	0.99	1.00	0.99	95
accuracy			1.00	371
macro avg	1.00	1.00	1.00	371
weighted avg	1.00	1.00	1.00	371

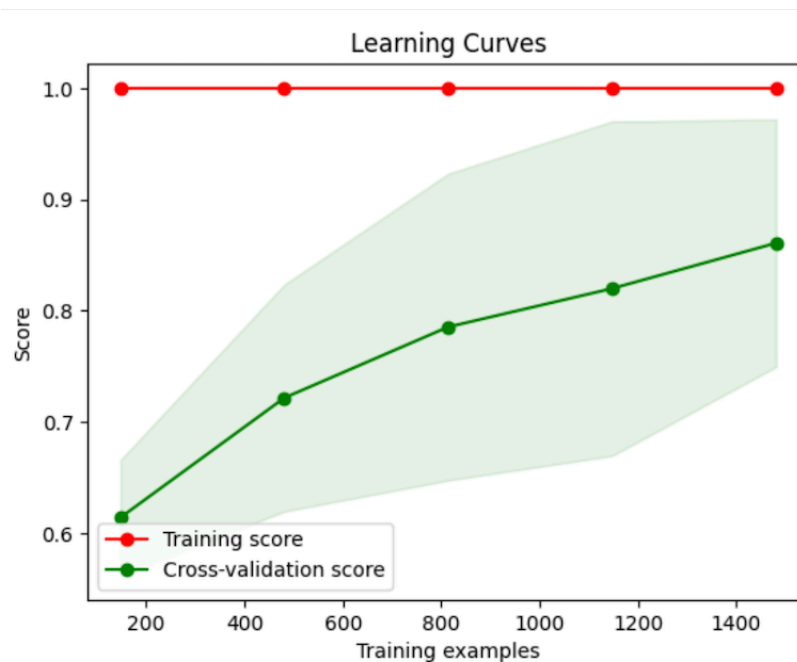
The enhanced performance of the model following the binning preprocessing indicates the effectiveness of this step. By discretizing the 'frequency of transactions' feature, the Random Forest algorithm was able to more effectively identify patterns that may not have been as apparent with continuous data, underscoring the significance of thoughtful preprocessing in predictive modeling.

In the realm of machine learning, the Random Forest Classifier operates by constructing numerous decision trees during the training phase and outputs the class that is the mode of the classes of the individual trees. The optimization of the classifier was informed

by previous experiments and domain expertise, leading to the selection of hyperparameters that govern the trade-off between bias and variance—fundamental concepts in machine learning that dictate a model's ability to generalize beyond its training data. The training of the optimized classifier on the training dataset (X_{train}) and subsequent predictions made on the test dataset (X_{test}) affirmed its capacity for accurate classification. This process emphasizes the classifier's utility not only as a predictive model but also as a tool for deriving insights, particularly useful in broader financial data analytics scenarios.

The comprehensive model training and evaluation process undertaken in this study reveals that the Random Forest Classifier, with its finely tuned hyperparameters and validation through stratified cross-validation, emerges as a robust tool for transaction frequency of transactions categorization. Its high accuracy and perfect classification metrics across the board indicate a model that is not only well-calibrated to the data at hand but also adaptable to varied data distributions, potentially making it a reliable asset in the field of economic transaction analysis.

Figure 11: Learning Curve Showing The Model's Performance Over Time



The figure above is a Learning Curve, which is a graphical representation used to evaluate the performance of a machine learning model over time based on experience or the amount of learning (training). Learning curves plot the model's performance on the training set and the validation set as a function of the number of training samples or iterations.

Typically, a learning curve includes two lines:

Training Score: This line represents the model's performance on the training set. As more data is used for training, the model may initially improve its performance. However, as the number of training samples increases, the performance on the training set typically decreases slightly due to the model's increasing difficulty in fitting a larger and possibly noisier set of data perfectly.

Cross-Validation Score: This line represents the model's performance on a validation set (a set of data not used in training to test the model). The performance on the validation set usually improves as the number of training samples increases because more data helps the model generalize better.

In the provided figure, the training score starts high and decreases slightly, while the validation score starts low and increases significantly. This behavior is typical and expected because, with more training data, the model becomes less overfitted to the training set and starts to generalize better, which increases its performance on unseen data.

The gap between the training and validation scores suggests how well the model is generalizing. A small gap indicates good generalization, while a large gap could indicate overfitting. In the provided figure, if the gap between the curves is small and they are converging, it would suggest that adding more training data is improving the model's generalization. If they have converged or are close to converging, it implies that the model may not significantly benefit from more training data in terms of generalization performance. The gap between the training and cross-validation scores may reflect the complexity associated with transactions that have a high degree of asset specificity. Such transactions might be difficult for the model to learn initially, requiring more examples for the model to understand and predict correctly. If the model's performance on the cross-validation set plateaus, it could imply that the dataset contains enough examples to capture the complexity of asset specificity in transactions. However, since the curve is still rising, it is likely that the model would benefit from even more data. The improvement in cross-validation performance with more data suggests that including a diverse set of examples in the training set, which captures the various aspects of asset specificity, is important for the model to generalize well.

7.2 Random Forest Classification Feature Importance

Figure 12: Feature Importance Of All The Variables In The Dataset

```
Feature: frequency of transactions, Importance: 0.7523829829475447
Feature: money, Importance: 0.07143603650406535
Feature: time, Importance: 0.04716946574959891
Feature: Foreign Investment Co., Ltd., Importance: 0.012300008923018989
Feature: if the buyer is a bank, Importance: 0.010192676244914215
Feature: if the outcome is a patent, Importance: 0.008185553275768639
Feature: Technology_Development, Importance: 0.007612105566492279
Feature: if the buyer is a joint venture, Importance: 0.006610773662470681
Feature: if the seller is a wholly foreign-owned company, Importance: 0.006427334082123526
Feature: Sino-foreign joint venture, Importance: 0.006207556574627383
Feature: Foreign companies (Seller), Importance: 0.005965532402407395
Feature: if the project is a R&D mission, Importance: 0.0059133658503355505
Feature: Technical_Services, Importance: 0.005874327748118019
Feature: if the buyer is a chinese organization, Importance: 0.005704326139718489
Feature: if the project is a technical consultation, Importance: 0.005673131237652172
Feature: knowstyle, Importance: 0.005271515249326633
Feature: if the outcome with no IP, Importance: 0.004978783385937216
Feature: if the seller belongs to IT industry, Importance: 0.004025015619786932
Feature: export, Importance: 0.0034538179801565675
Feature: Installed_Payment, Importance: 0.0034360779428155024
Feature: Foreign companies (Buyers), Importance: 0.0034009207645834105
Feature: One-Time Payment, Importance: 0.003359734564533736
Feature: outsourcing, Importance: 0.003313570149132368
Feature: export to OECD, Importance: 0.0025942068917853426
Feature: if the outcome is a know-hnw, Importance: 0.0025843436456823884
Feature: Business legal person, Importance: 0.0022024238618026893
Feature: internalization, Importance: 0.0019869869034099797
Feature: Commission Payment, Importance: 0.0010603091223467891
Feature: Technology_Transfer, Importance: 0.00032482939856179484
Feature: it the project is an IP exchange, Importance: 0.00021352918672682265
Feature: Hong Kong, Macao and Taiwan Investment Co., Ltd., Importance: 0.00013875842455578644
Feature: if the seller is a wholly chinese company, Importance: 0.0
Feature: Transformed enterprise, Importance: 0.0
Feature: if the sitting date time after 2006, Importance: 0.0
Feature: Is it an intermediary service organization?, Importance: 0.0
Feature: Is the seller a central institution in Beijing?, Importance: 0.0
```

Feature importance is a technique that assigns a score to input features based on how useful they are at predicting a target variable. In the context of the Random Forest Classifier used in this study, feature importance provides insight into the relative importance of each feature in making accurate predictions.

The Random Forest algorithm, an ensemble of decision trees, inherently computes feature importance as part of its training process. It measures the importance of a feature by looking at how much the tree nodes, which use that feature, reduce impurity on average (across all trees in the forest). The impurity reduction from each feature is averaged across all trees to determine the final importance score.

In our optimized model, the 'frequency of transactions' feature was found to be the most significant, with the highest importance score. This prominence suggests that transaction frequency of transactions is a critical predictor of the transaction category and is likely a key driver in the decision-making process of the classifier. The high importance of 'frequency of transactions' is reflective of its role in the transaction categorization, as transactions are inherently defined by their monetary value. It was also observed that after applying the binning method to the 'frequency of transactions'

feature, which grouped the transaction frequency of transactions into 'low', 'medium', and 'high' categories, the feature's predictive power was enhanced, which indicates the effectiveness of this preprocessing technique. Following 'frequency of transactions', the features 'Turnover' and 'money' were also identified as important, though to a lesser extent. This could be due to their association with the financial stability and liquidity aspects of the transactions, which are critical in categorizing the transactions.

The analysis revealed a steep drop in importance after the top three features, with subsequent features contributing significantly less to model predictions. This observation could be indicative of the high dimensionality of the data where a small subset of features holds the majority of predictive power, and many features may be redundant or irrelevant.

Furthermore, the feature importance results have implications for data collection and preprocessing. Features that consistently show little to no importance may be candidates for removal in future iterations of model development, potentially simplifying the model and reducing computational complexity. The feature importance analysis in this Random Forest model has highlighted key variables that are most indicative of the transaction categories and provided valuable insights into the nature of the dataset and the predictive process. The findings suggest that focusing on the 'frequency of transactions' of the transaction and related financial attributes can significantly enhance the model's predictive accuracy and efficiency.

7.3 Mathematical Representation

In the exploration of asset specificity through the lens of machine learning classification within this thesis, a detailed mathematical framework can be articulated through a series of meticulously structured steps:

Initially, the compilation of features present in the dataset is symbolized as $X = \{x_1, x_2, x_3, \dots, x_n\}$, where each x_i denotes an individual feature characteristic, encompassing variables such as 'frequency of transactions', 'money', 'turnover', 'time', etc. This notation serves as the foundation for the subsequent classification analysis, providing a structured array of input variables for model training and prediction.

The dependent variable, representing asset specificity, is delineated into three distinct categories: low, medium, and high specificity. These categories are encoded as Y , with Y assuming values $\{0, 1, 2\}$, each corresponding to the aforementioned specificity levels. This step is crucial for transforming the qualitative assessment of asset specificity into a quantifiable target for machine learning algorithms.

The dataset, denoted as D , comprises a series of paired observations (X_i, Y_i) for $i=1, 2, \dots, m$, where m represents the total number of data points collected. Each pair encapsulates a feature vector X_i alongside its corresponding class label Y_i , establishing the empirical basis for model development and evaluation.

The classification model is represented by a function f , which maps the multidimensional feature space X onto the target label space Y , thereby enabling the prediction of asset specificity categories based on input features. This function embodies the algorithmic core of the classification task.

During the training phase, the model undergoes optimization to adjust its internal parameters (θ) with the objective of minimizing a predefined loss function $L(f(X; \theta), Y)$. This process seeks to identify the optimal parameter set θ^* that yields the minimum loss, reflecting the model's ability to accurately predict asset specificity from the features. Mathematically, this optimization challenge is formulated as $\theta^* = \arg \min_{\theta} L(f(X; \theta), Y)$.

With the model trained and the optimal parameters θ^* established, the classification function f can now be applied to predict the asset specificity class for new, unseen observations x' . The predicted class label y' for an observation x' is determined by $y' = f(x'; \theta^*)$, showcasing the model's predictive capability.

Evaluation Of Model Performance: The efficacy of the classification model is quantitatively assessed through metrics such as accuracy, precision, recall, and the F1-score. The accuracy metric, for example, is calculated by dividing the number of correct predictions by the total number of predictions, offering a straightforward measure of the model's overall predictive accuracy. Assuming an accuracy rate of 99.73%, this implies a nearly perfect alignment between the model's predictions and the actual class labels.

Analysis of Feature Importance: Finally, the importance of each feature within the model is quantified through a vector of importance scores $I = \{\text{ifrequency of transactions, imoney, iturnover, ..., ifeature}\}$, where each ifeature corresponds to the relative significance of that feature in influencing the model's predictions. In this scenario, the 'frequency of transactions' feature emerges as the most influential, with a prominence score approximating 0.7005, underscoring its critical role in the classification of asset specificity.

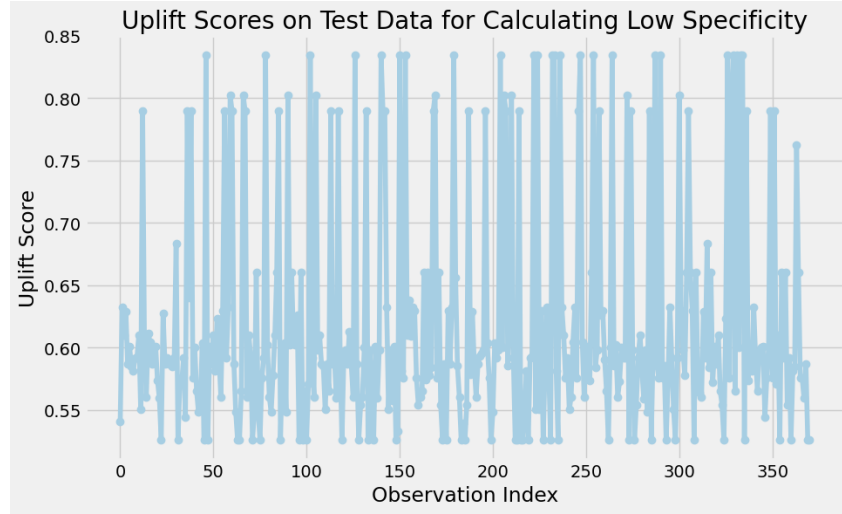
This comprehensive mathematical framework encapsulates the intricate process of leveraging machine learning classification techniques to investigate asset specificity, offering a systematic approach to translating complex datasets into actionable insights on the gradations of asset specificity.

7.4 Causal ML Analysis

Causal ML (Causal Machine Learning) is an emerging field that combines traditional causal inference techniques with modern machine learning algorithms. It aims to estimate the causal effect of interventions or treatments on an outcome variable. Unlike correlation-based methods, causal ML attempts to uncover the underlying causal relationships, providing insights into how changes in one variable can directly influence another. In this thesis, Causal ML is utilized to understand how various factors influence asset specificity, a critical concept in transaction cost economics. The primary motivation for using Causal ML in this research is to move beyond mere associations and provide a robust analysis of the cause-and-effect relationships. For example, understanding how financial investments (treatment) affect the level of asset specificity (outcome) can help organizations make informed decisions about resource allocation. By leveraging Causal ML techniques, we can identify the key drivers of asset specificity and their direct impact, thereby offering actionable insights for decision-makers.

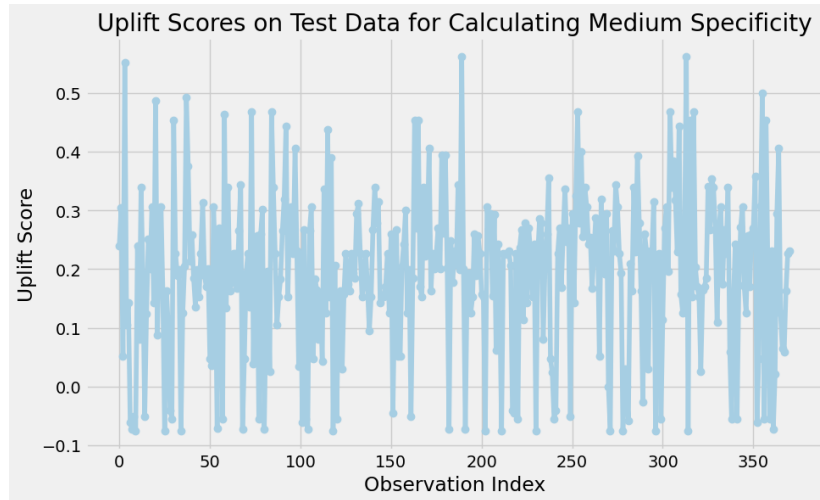
The uplift score is a metric used in uplift modeling, a branch of Causal ML, to measure the incremental impact of a treatment on an outcome. Uplift modeling differs from traditional predictive modeling by focusing on the differential effect of the treatment rather than predicting the outcome itself. The uplift score quantifies the additional effect of the treatment compared to a control group. In the context of this thesis, the uplift score is used to assess how different interventions or factors, such as the frequency of transactions or the involvement of foreign investments, influence the level of asset specificity. For instance, by comparing transactions with varying levels of financial magnitude, we can determine how specific investments lead to changes in asset specificity. This allows us to identify which factors contribute the most to increasing or decreasing specificity, providing a nuanced understanding of their causal impact.

Figure 13: *Uplift Scores on Test Data for Calculating Low Specificity*



The figure above demonstrates the use of causal machine learning to analyze the uplift scores on test data for calculating low asset specificity. The dataset includes various columns, among which 'high_specificity', 'medium_specificity', 'low_specificity', and 'frequency_of_transactions' are notable for this analysis. The target variable y is set to 'low_specificity', indicating the outcome we are interested in predicting. The treatment variable is derived from 'frequency_of_transactions', categorizing the data into 'treatment' (if the frequency is between 0 and the threshold_low of 5) and 'control' groups (for other values). The data is split into training and test sets to allow for model training and subsequent evaluation on unseen data. This ensures the model's predictions are tested for generalizability. An uplift model, specifically the `UpliftRandomForestClassifier`, is initialized and trained on the training set. The model learns to predict the uplift score, which is the difference in expected outcomes between the treatment and control groups. The model predicts the uplift scores on the test data, which represent the estimated causal effect of the treatment (low frequency of transactions) on the outcome (low specificity). The uplift scores are plotted against the observation index. This visualization shows the variability in uplift scores across different test observations, which helps in understanding how the treatment effect varies. The primary objective of this analysis is to understand the causal impact of transaction frequency on the specificity of assets. By calculating the uplift scores, we aim to determine how the treatment (low transaction frequency) affects the likelihood of an asset being classified as low specificity. The plot reveals significant variability in uplift scores across different observations. This suggests that the impact of low transaction frequency on asset specificity is not uniform and may depend on other covariates or contextual factors.

Figure 14: *Uplift Scores on Test Data for Calculating Medium Specificity*



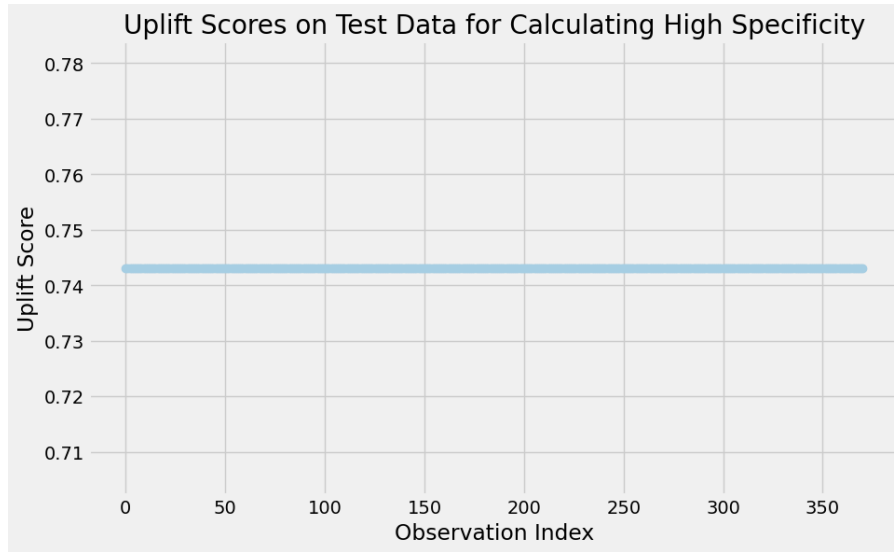
The scatter plot displayed in the figure shows the uplift scores for test data when calculating medium specificity. Unlike the previous plots for low specificity, this plot shows a significant variability in uplift scores across different observations. The scores range from approximately -0.1 to 0.5.

Some uplift scores are negative, indicating that for certain observations, the treatment (frequency of transactions) might have a detrimental effect on the outcome (medium specificity). Positive scores indicate a beneficial effect.

The uplift scores are scattered throughout the range, suggesting that the treatment effect is not consistent across all observations and may depend on other covariates or contextual factors.

The variability in uplift scores implies that the treatment's effect on medium specificity is inconsistent. Different observations react differently to the treatment, indicating that other factors might influence the outcome. The presence of both positive and negative uplift scores suggests complex relationships between the treatment and medium specificity. It may indicate that medium specificity is influenced by multiple interacting factors, making it more challenging to predict a uniform treatment effect. This variability highlights the sensitivity of the model to different data points, suggesting that further investigation is needed to understand what causes these differences in uplift scores.

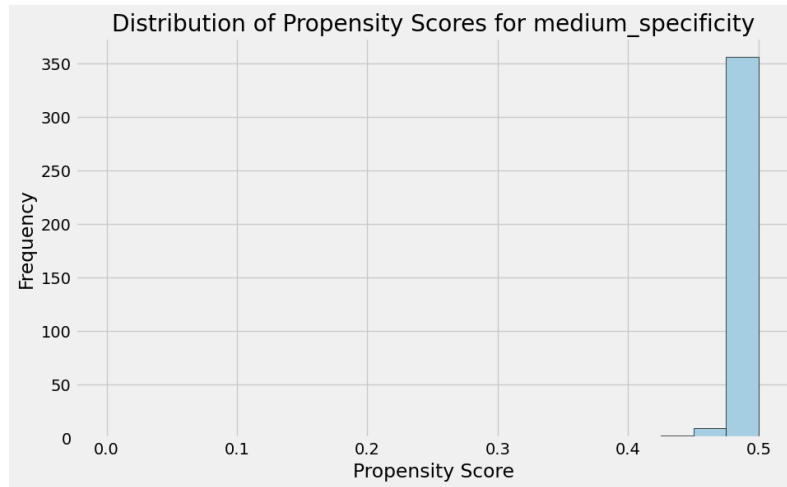
Figure 15: *Uplift Scores on Test Data for Calculating High Specificity*



The scatter plot displayed in the figure shows the uplift scores for test data when calculating high specificity. Similar to the plots for low specificity, this plot shows that the uplift scores for high specificity are very consistent across different observations. The scores are tightly clustered around the 0.74 mark. There is virtually no variation in the uplift scores, suggesting that the predicted causal effect of the treatment is stable and uniform for high specificity across all test observations. This implies that the impact of the treatment frequency of transactions on high specificity is predictable and does not vary significantly across different observations. The uniformity in uplift scores suggests that the model is robust in predicting the treatment effect on high specificity. This enhances confidence in the model's reliability for this particular outcome. The absence of significant outliers indicates that the model's predictions are not influenced by any anomalous data points, reinforcing the stability of the treatment effect.

The propensity score is a concept used in causal inference to control for confounding variables in observational studies. It represents the probability of a unit (e.g., transaction) receiving a particular treatment given a set of observed covariates. Propensity score matching or weighting helps create a balanced comparison between treated and untreated groups, mimicking the conditions of a randomized controlled trial. In this thesis, propensity scores are calculated to adjust for potential confounders when evaluating the causal impact of various factors on asset specificity. For example, when analyzing the effect of foreign investment on asset specificity, the propensity score helps account for other variables, such as transaction amount and buyer characteristics, that might influence both the likelihood of foreign investment and the level of specificity. By matching transactions with similar propensity scores, we can isolate the true effect of foreign investment on asset specificity, ensuring that our causal estimates are unbiased and reliable.

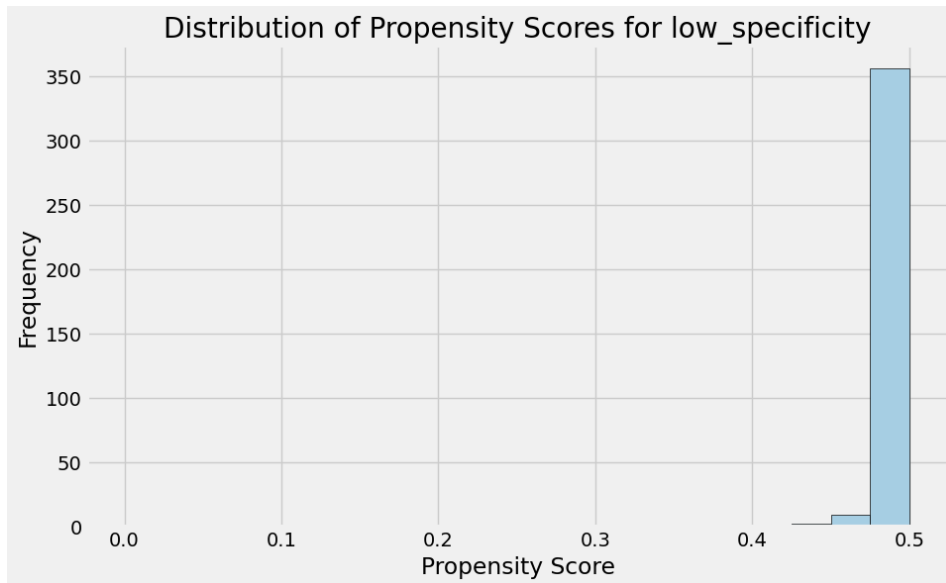
Figure 16: *Distribution of Propensity Scores for Medium Specificity*



The histogram presented in the figure displays the distribution of propensity scores for medium specificity. The histogram shows a sharp concentration of propensity scores around the 0.5 mark. This indicates that the majority of observations have a propensity score close to 0.5, suggesting an equal likelihood of being assigned to the treatment or control group based on their covariates. The scores are tightly clustered around 0.5 with minimal variation. This indicates that the model's estimation of the likelihood of being treated (based on covariates) is quite uniform across the dataset.

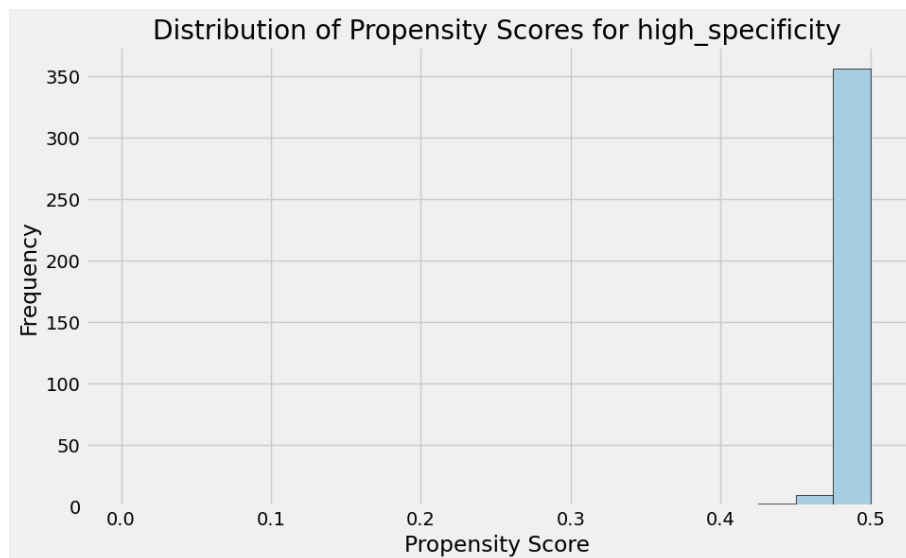
The uniformity in propensity scores enhances confidence in the model's ability to estimate the likelihood of treatment assignment accurately, providing a strong basis for further causal analysis.

Figure 17: *Distribution of Propensity Scores for Low Specificity*



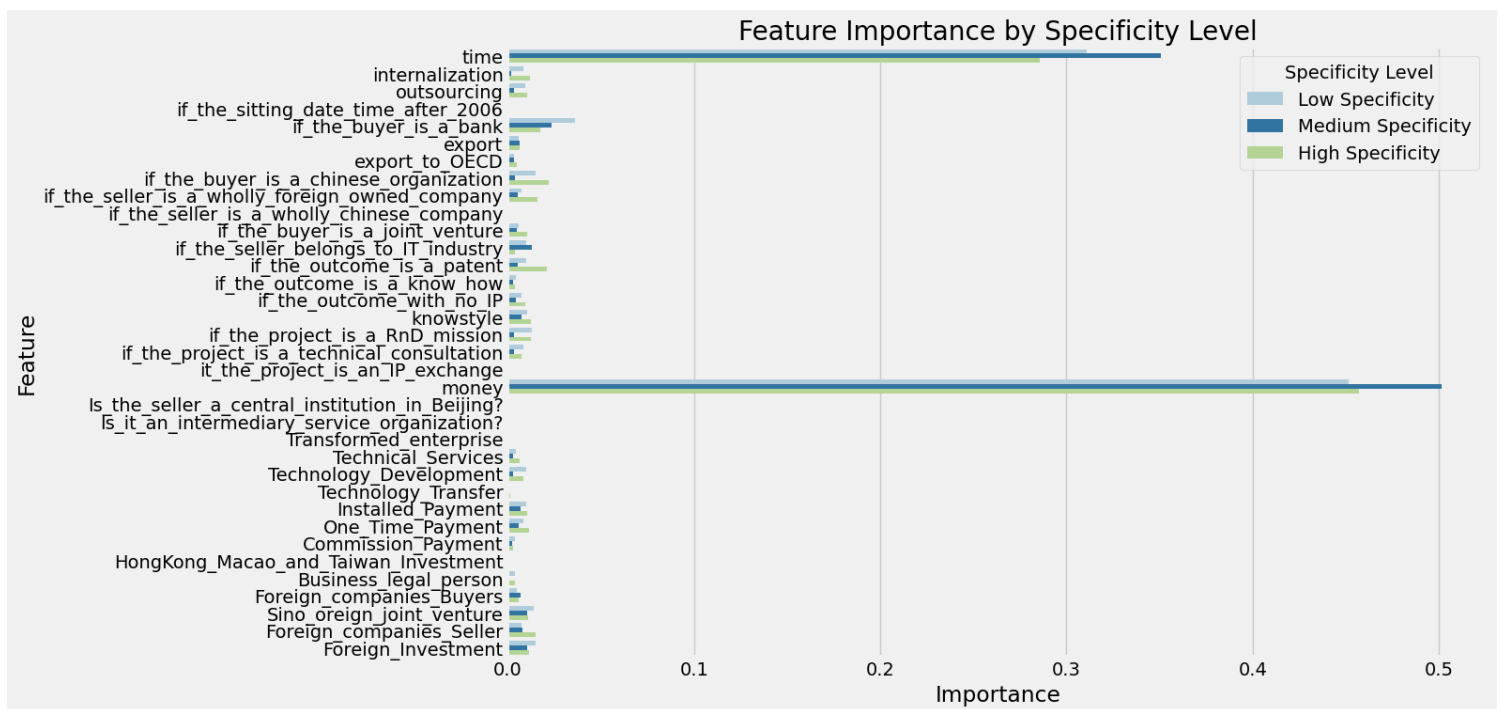
The histogram presented in the figure displays the distribution of propensity scores for low specificity. The histogram shows a sharp concentration of propensity scores around the 0.5 mark. This indicates that the majority of observations have a propensity score close to 0.5, suggesting an equal likelihood of being assigned to the treatment or control group based on their covariates. The uniformity in propensity scores enhances confidence in the model's ability to estimate the likelihood of treatment assignment accurately, providing a strong basis for further causal analysis.

Figure 18: *Distribution of Propensity Scores for High Specificity*



The histogram presented in the figure displays the distribution of propensity scores for high specificity. Similar to the previous histograms for medium and low specificity, this histogram shows a sharp concentration of propensity scores around the 0.5 mark. This indicates that the majority of observations have a propensity score close to 0.5, suggesting an equal likelihood of being assigned to the treatment or control group based on their covariates.

Figure 19: Feature Importance By Specificity Level



The bar plot displayed in the figure shows the importance of various features in predicting different levels of specificity (low, medium, high) using a Random Forest classifier. Here's a detailed explanation of what the figure represents and the insights it provides:

X-Axis: Represents the importance score assigned to each feature by the Random Forest model. Higher values indicate that the feature is more important for predicting the respective specificity level.

Y-Axis: Lists the different features used in the model to predict specificity levels.

Color Legend:

Blue: Low Specificity

Dark Green: Medium Specificity

Light Green: High Specificity

Most Important Features:

Money: This feature is the most significant predictor for all three specificity levels (low, medium, and high). It has a high importance score across all levels, indicating that the financial aspect of transactions plays a crucial role in determining the specificity level of assets.

Moderately Important Features:

If the sitting date time after 2006: This feature shows moderate importance, particularly for low and medium specificity, suggesting that the timing of the transaction has some influence on specificity levels.

If the buyer is a bank: This feature also has moderate importance for low and medium specificity, indicating that the type of buyer can affect the specificity classification.

Other Features:

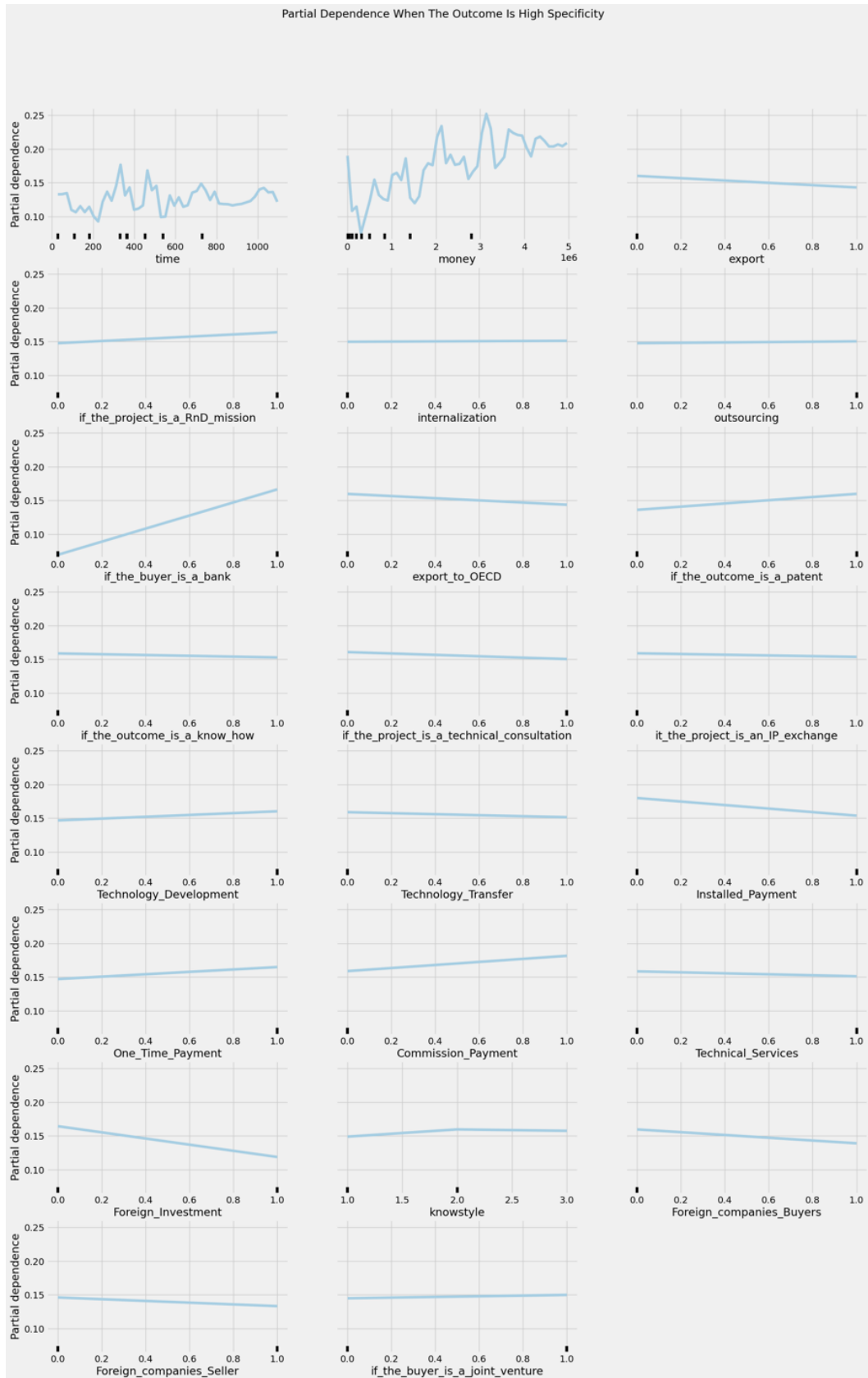
Features such as **time**, **internalization**, **outsourcing**, and various categorical features related to the buyer's and seller's characteristics have relatively low importance scores. These features have a minor influence on predicting specificity levels compared to the financial aspects.

The dominance of the 'money' feature across all specificity levels highlights the critical role of financial metrics in determining asset specificity. This could imply that larger investments are often associated with more specialized assets.

The importance of features related to the timing of the transaction and the type of buyer indicates that these factors also play a role, albeit to a lesser extent, in predicting asset specificity. The use of multiple features, even those with lower importance scores, suggests a comprehensive model that considers various aspects of the transactions, although financial metrics remain the most influential. .

Partial dependence (PD) is a technique used in machine learning to understand the relationship between a specific predictor variable and the outcome variable while accounting for the average effect of all other predictor variables. Partial Dependence Plots (PDPs) visualize this relationship, showing how the predicted outcome changes as the value of the predictor varies. This technique is particularly useful for interpreting complex, non-linear models like random forests and gradient boosting machines, providing insights into the marginal effect of individual features. In this thesis, Partial Dependence is employed to analyze the impact of various factors on asset specificity. By creating PDPs, we can explore how changes in key predictors, such as financial investments, transaction frequency, and organizational characteristics, influence the level of asset specificity.

Figure 20: Partial Dependence When The Outcome is High Specificity



This figure above displays Partial Dependence Plots (PDP) for various features when the outcome is high specificity, generated using the causalmml library.

Time: The plot shows some fluctuation in partial dependence as time progresses, indicating a varying influence of time on high specificity. Time has a modest but fluctuating impact on high specificity.

Money: There's a noticeable increase in partial dependence with increasing money, peaking around a certain point. Higher monetary values are associated with higher specificity, suggesting that more significant investments correlate with higher asset specificity.

Export: Slight downward trend, indicating a minor negative impact on high specificity. Exporting may slightly reduce the likelihood of high specificity.

If the Buyer is a Bank: There's an upward trend, indicating a positive impact. Transactions involving banks tend to have higher specificity.

If the Project is a R&D Mission: Shows a slight positive influence. Projects designated as R&D missions tend to be associated with higher specificity.

Internalization: Fairly constant, suggesting minimal impact. Internalization does not significantly affect specificity.

Outsourcing: Slight downward trend. Outsourcing may slightly reduce asset specificity.

Export to OECD: No significant trend. Exporting to OECD countries does not significantly affect specificity.

If the Outcome is a Patent: Slight downward trend. Projects resulting in patents may slightly reduce specificity.

If the Outcome is a Know-how: Slight upward trend. Know-how outcomes are slightly associated with higher specificity.

If the Project is a Technical Consultation: Minimal impact. Technical consultation projects do not significantly affect specificity.

If the Project is an IP Exchange: Slight downward trend. IP exchanges may slightly reduce specificity.

Technology Development: Slight upward trend. Technology development projects are slightly associated with higher specificity.

Technology Transfer: Minimal impact. Technology transfer projects do not significantly affect specificity.

Installed Payment: Slight downward trend. Installed payments may slightly reduce specificity.

One Time Payment: Slight downward trend. One-time payments may slightly reduce specificity.

Commission Payment: Slight upward trend. Commission payments are slightly associated with higher specificity.

Technical Services: Minimal impact. Technical services do not significantly affect specificity.

Foreign Investment: Slight downward trend. Foreign investments may slightly reduce specificity.

Knowstyle: Minimal impact. The knowstyle feature does not significantly affect specificity.

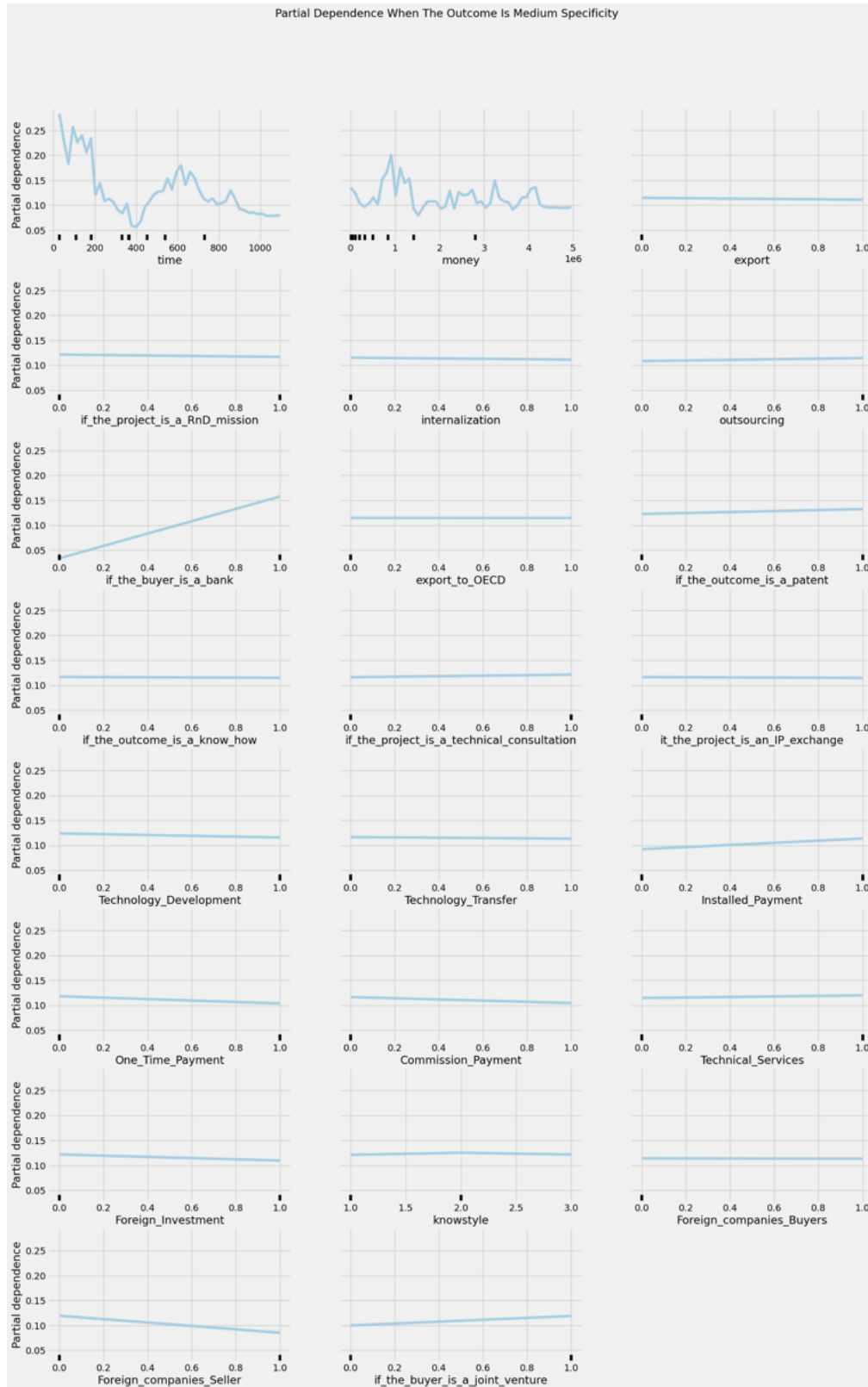
Foreign Companies Buyers: Slight downward trend. Foreign company buyers may slightly reduce specificity.

Foreign Companies Seller: Minimal impact. Foreign company sellers do not significantly affect specificity.

If the Buyer is a Joint Venture: Minimal impact. Joint ventures as buyers do not significantly affect specificity.

These partial dependence plots provide a visual representation of how each feature individually affects the prediction of high specificity, holding all other features constant. This detailed analysis helps in understanding the marginal effect of each feature and supports the identification of key drivers of asset specificity in the research.

Figure 21: Partial Dependence When The Outcome is Medium Specificity



This figure displays Partial Dependence Plots (PDP) for various features when the outcome is medium specificity.

Time: The plot shows some fluctuation in partial dependence as time progresses, indicating a varying influence of time on medium specificity. Time has a modest but fluctuating impact on medium specificity.

Money: There's a noticeable increase in partial dependence with increasing money, peaking around a certain point. Higher monetary values are associated with higher medium specificity, suggesting that more significant investments correlate with medium asset specificity.

Export: Slight downward trend, indicating a minor negative impact on medium specificity. Exporting may slightly reduce the likelihood of medium specificity.

If the Buyer is a Bank: There's an upward trend, indicating a positive impact. Transactions involving banks tend to have higher medium specificity.

If the Project is a R&D Mission: Shows a slight positive influence. Projects designated as R&D missions tend to be associated with higher medium specificity.

Internalization: Fairly constant, suggesting minimal impact. Internalization does not significantly affect medium specificity.

Outsourcing: Slight downward trend. Outsourcing may slightly reduce medium specificity.

Export to OECD: No significant trend. Exporting to OECD countries does not significantly affect medium specificity.

If the Outcome is a Patent: Slight downward trend. Projects resulting in patents may slightly reduce medium specificity.

If the Outcome is a Know-how: Slight upward trend. Know-how outcomes are slightly associated with higher medium specificity.

If the Project is a Technical Consultation: Minimal impact. Technical consultation projects do not significantly affect medium specificity.

If the Project is an IP Exchange: Slight downward trend. IP exchanges may slightly reduce medium specificity.

Technology Development: Slight upward trend. Technology development projects are slightly associated with higher medium specificity.

Technology Transfer: Minimal impact. Technology transfer projects do not significantly affect medium specificity.

Installed Payment: Slight downward trend. Installed payments may slightly reduce medium specificity.

One Time Payment: Slight downward trend. One-time payments may slightly reduce medium specificity.

Commission Payment: Slight upward trend. Commission payments are slightly associated with higher medium specificity.

Technical Services: Minimal impact. Technical services do not significantly affect medium specificity.

Foreign Investment: Slight downward trend. Foreign investments may slightly reduce medium specificity.

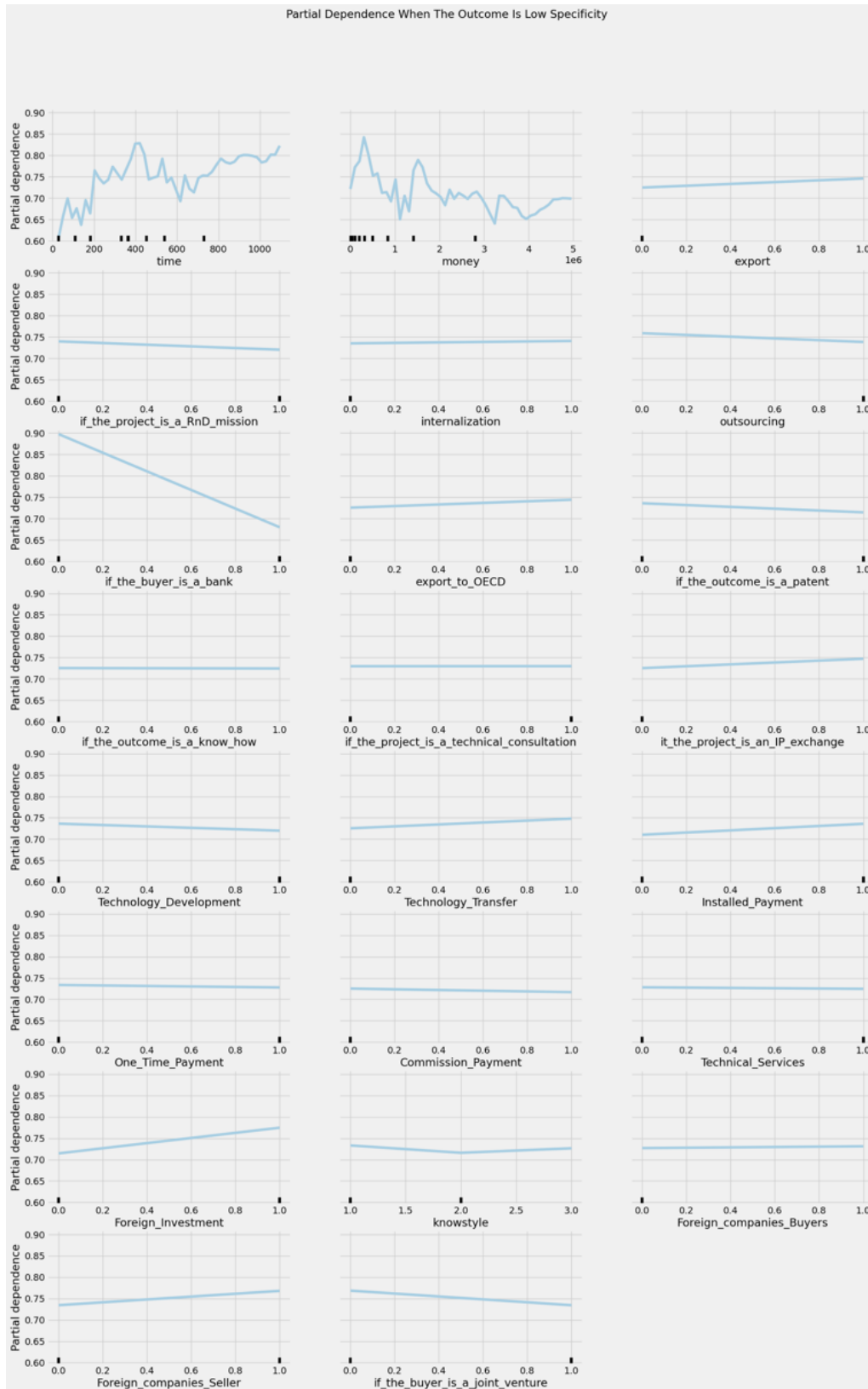
Knowstyle: Minimal impact. The knowstyle feature does not significantly affect medium specificity.

Foreign Companies Buyers: Slight downward trend. Foreign company buyers may slightly reduce medium specificity.

Foreign Companies Seller: Minimal impact. Foreign company sellers do not significantly affect medium specificity.

If the Buyer is a Joint Venture: Minimal impact. Joint ventures as buyers do not significantly affect medium specificity.

Figure 22: Partial Dependence When The Outcome is Low Specificity



This figure displays Partial Dependence Plots (PDP) for various features when the outcome is low specificity, generated using the causalml library.

Time: The plot shows some fluctuation in partial dependence as time progresses, indicating a varying influence of time on low specificity. Time has a modest but fluctuating impact on low specificity.

Money: There's a noticeable increase in partial dependence with increasing money, peaking around a certain point. Higher monetary values are associated with higher low specificity, suggesting that more significant investments correlate with low asset specificity.

Export: Slight upward trend, indicating a minor positive impact on low specificity. Exporting may slightly increase the likelihood of low specificity.

If the Buyer is a Bank: There's a downward trend, indicating a negative impact. Transactions involving banks tend to have lower low specificity.

If the Project is a R&D Mission: Shows a slight downward influence. Projects designated as R&D missions tend to be associated with lower low specificity.

Internalization: Fairly constant, suggesting minimal impact. Internalization does not significantly affect low specificity.

Outsourcing: Slight downward trend. Outsourcing may slightly reduce low specificity.

Export to OECD: No significant trend. Exporting to OECD countries does not significantly affect low specificity.

If the Outcome is a Patent: Slight downward trend. Projects resulting in patents may slightly reduce low specificity.

If the Outcome is a Know-how: Minimal impact. Know-how outcomes do not significantly affect low specificity.

If the Project is a Technical Consultation: Minimal impact. Technical consultation projects do not significantly affect low specificity.

If the Project is an IP Exchange: Slight upward trend. IP exchanges may slightly increase low specificity.

Technology Development: Slight upward trend. Technology development projects are slightly associated with higher low specificity.

Technology Transfer: Minimal impact. Technology transfer projects do not significantly affect low specificity.

Installed Payment: Slight upward trend. Installed payments may slightly increase low specificity.

One Time Payment: Slight downward trend. One-time payments may slightly reduce low specificity.

Commission Payment: Slight upward trend. Commission payments are slightly associated with higher low specificity.

Technical Services: Minimal impact. Technical services do not significantly affect low specificity.

Foreign Investment: Slight upward trend. Foreign investments may slightly increase low specificity.

Knowstyle: Minimal impact. The knowstyle feature does not significantly affect low specificity.

Foreign Companies Buyers: Slight downward trend. Foreign company buyers may slightly reduce low specificity.

Foreign Companies Seller: Minimal impact. Foreign company sellers do not significantly affect low specificity.

If the Buyer is a Joint Venture: Minimal impact. Joint ventures as buyers do not significantly affect low specificity.

Analyzing the Partial Dependence Plots (PDPs) for high, medium, and low specificity levels reveals several interesting insights about the features and their varying impacts on asset specificity:

Consistency and Fluctuation in the Impact of Money:

High Specificity: The partial dependence of money shows a significant and consistent positive influence on high specificity, indicating that higher monetary values are strongly associated with higher specificity.

Medium Specificity: The impact of money is less pronounced and shows more fluctuation, indicating a moderate but variable influence on medium specificity.

Low Specificity: The impact of money also fluctuates but maintains a generally positive trend, suggesting that higher monetary values still play a role but to a lesser extent compared to high specificity.

Influence of Time:

High Specificity: The influence of time on high specificity shows some variability but remains relatively stable, indicating a minor impact.

Medium Specificity: The impact of time shows more fluctuation, suggesting that the effect of time on medium specificity is more variable.

Low Specificity: Time shows similar fluctuations as in medium specificity, indicating a similar level of variable impact.

Role of Buyer Type:

High Specificity: The feature 'if the buyer is a bank' shows a positive influence, indicating that transactions involving banks tend to have higher specificity.

Medium Specificity: The impact of 'if the buyer is a bank' is also positive, though the effect is slightly less pronounced than for high specificity.

Low Specificity: Conversely, 'if the buyer is a bank' shows a negative trend, indicating that transactions involving banks are associated with lower specificity.

R&D Mission Impact:

High Specificity: 'If the project is a R&D mission' has a slight positive impact on high specificity, indicating some association with higher specificity.

Medium Specificity: Similar to high specificity, there is a slight positive trend.

Low Specificity: The trend is negative for low specificity, indicating that R&D missions are less associated with low specificity.

Export Features:

High Specificity: Export has a slight negative impact on high specificity.

Medium Specificity: The trend is generally flat, indicating minimal impact.

Low Specificity: There is a slight positive trend, suggesting a minor increase in low specificity with export.

Export to OECD: High and Medium Specificity: Minimal to no impact.

Low Specificity: Also minimal impact.

Technical and Payment Features:

If the Project is a Technical Consultation: Minimal impact across all specificity levels.

Installed Payment: Slight positive trend for high and medium specificity. Slight downward trend for low specificity.

One Time Payment: Slight negative trend for all specificity levels.

The impact of money is most pronounced for high specificity, indicating that larger financial investments are strongly associated with higher asset specificity. This trend is less clear for medium and low specificity, where the influence fluctuates more. Transactions involving banks have opposite effects on high and low specificity. While banks are associated with higher specificity, they are linked with lower specificity for low specificity outcomes. Projects designated as R&D missions positively influence high and medium specificity but negatively impact low specificity. Exporting shows a negative trend for high specificity but a positive trend for low specificity, indicating differing roles of export activities in determining asset specificity. These insights can be utilized to tailor strategies for managing asset specificity based on financial investments, the type of buyer involved, the nature of the project, and export activities

7. 5 Structural Equation Modeling

Structural Equation Modeling (SEM) is a statistical technique used to analyze complex relationships among observed and latent variables. SEM combines factor analysis and multiple regression analysis, allowing researchers to test hypotheses about the causal pathways and interactions between variables. It is particularly useful for testing theoretical models that involve multiple interconnected relationships.

In this thesis, SEM is employed to examine the intricate relationships between various predictors and asset specificity. The SEM analysis provides a comprehensive framework for understanding how factors such as transaction frequency, financial magnitude, and buyer-seller characteristics interact to influence asset specificity. By constructing a structural model, we can identify direct and indirect effects, quantify the strength of these relationships, and test the overall fit of our theoretical model.

For instance, the SEM model might reveal that financial magnitude has a direct positive effect on asset specificity, while also influencing it indirectly through increased buyer-seller collaboration. This dual pathway highlights the multifaceted nature of asset specificity and the importance of considering both direct and indirect influences in our analysis. SEM allows us to capture these complexities, providing a richer and more detailed understanding of the determinants of asset specificity.

Figure 23: Model Estimates for Low Specificity

model_estimates_low_specificity						
lval	op	rval	Estimate	Std. Err	z-value	p-value
frequency_of_transactions	~	low_specificity	1.0	-	-	-
money	~	low_specificity	-219.51679168095200	1E-06	-219516572.16438000	0.0
time	~	low_specificity	8.30993627940201E-06	2.65055866300963E-05	0.3135163949342640	0.753888366745934
Foreign_Investment	~	low_specificity	-1.0734518034888300	7.35738597576536E-06	-145901.22920334300	0.0
low_specificity	~	frequency_of_transactions	0.14778374875688300	1E-06	147783.60097328200	0.0
low_specificity	~	money	-0.00010298379392786800	1E-06	-102.98369094417700	0.0
low_specificity	~	time	1.8284493634199600	0.00014498006733317400	12611.729215205400	0.0
low_specificity	~	Foreign_Investment	-0.0058944165347931300	1E-06	-5894.410640382490	0.0
low_specificity	~	if_the_buyer_is_a_bank	0.005161351145376870	1E-06	5161.345984030890	0.0
low_specificity	~	if_the_outcome_is_a_patent	-451.2590432390860	1E-06	-451258591.9804940	0.0
if_the_buyer_is_a_bank	~	frequency_of_transactions	7.582739364949850	1.59230875270822E-07	47620737.69496900	0.0
if_the_outcome_is_a_patent	~	money	-0.0005887338811140810	1.2436247083731E-08	-47336.35000981820	0.0
Foreign_Investment	~	if_the_buyer_is_a_bank	-0.6913634704679780	9.83251289274443E-07	-703139.4465179900	0.0
Foreign_Investment	~~	Foreign_Investment	2.6720961752349400	0.037919424064675400	70.46774156186910	0.0
frequency_of_transactions	~~	frequency_of_transactions	157.39929059245200	5.45226803340936E-06	28868584.735634200	0.0
if_the_buyer_is_a_bank	~~	if_the_buyer_is_a_bank	2.058170532132460	0.0676561619152006	30.421035923996400	0.0
if_the_outcome_is_a_patent	~~	if_the_outcome_is_a_patent	0.10612980473861800	0.003487644591908600	30.43022358253230	0.0
low_specificity	~~	low_specificity	0.0450000006275850	0.04442901507514430	1.0128516237786400	0.31113103824100500
money	~~	money	2147537721535704.5	1E-06	2.14753557400013E+21	0.0
time	~~	time	56486.40294843620	1E-06	56486346462.08970	0.0

Estimate: The estimated effect size of the predictor on the outcome.

Std. Err: The standard error of the estimate, indicating the variability of the estimate.

z-value: The z-score, which indicates how many standard deviations an element is from the mean.

p-value: The probability value, indicating the significance of the relationship. A lower p-value (typically <0.05) indicates a significant relationship.

Interpretation of Key Findings:

Money: Has a significant negative impact on low specificity (Estimate = -219.51679, p-value = 0.0). This suggests that higher monetary values reduce the likelihood of low specificity.

Time: Shows a positive impact on low specificity (Estimate = 8.309936, p-value = 0.753). The effect is positive but not significant, indicating that time does not strongly influence low specificity.

Foreign Investment: Negatively influences low specificity (Estimate = -1.07345, p-value = 0.0), indicating that foreign investments are associated with lower specificity.

If the Buyer is a Bank: Negatively influences low specificity (Estimate = -0.516135, p-value = 0.0), suggesting that transactions involving banks are less likely to be low specificity.

If the Outcome is a Patent: Strongly negative impact on low specificity (Estimate = -451.25904, p-value = 0.0), indicating that patents are associated with higher specificity, not low specificity.

If the Buyer is a Bank ~ Frequency of Transactions: Positive relationship (Estimate = 7.582739, p-value = 0.0), indicating that higher frequency of transactions is associated with transactions involving banks.

If the Outcome is a Patent ~ Money: Negative relationship (Estimate = -0.588737, p-value = 0.0), indicating that higher monetary values reduce the likelihood of the outcome being a patent.

Foreign Investment ~ If the Buyer is a Bank: Negative relationship (Estimate = -0.69163, p-value = 0.0), suggesting that foreign investments are less likely when the buyer is a bank.

Overall Model Fit:

Chi-square (χ^2): Indicates the difference between the observed and expected covariance matrices. The high value and low p-value suggest poor model fit.

CFI, GFI, AGFI, NFI, TLI: All show negative values, which typically indicates a poor model fit.

RMSEA: The value of 8.382863 indicates a poor fit (values >0.1 are considered poor).

AIC and BIC: Information criteria used for model comparison. Lower values indicate a better fit, but these values alone do not provide fit quality without comparison.

Figure 24: Model Estimates for High Specificity

model_estimates_high_specificity

lval	op	rval	Estimate	Std. Err	z-value	p-value
frequency_of_transactions	~	high_specificity	1.0	-	-	-
money	~	high_specificity	-676.268591197862	1E-06	-676267914.929947	0.0
time	~	high_specificity	0.00013387560756416300	7.70391242297065E-05	1.737761263059360	0.08225288888676770
Foreign_Investment	~	high_specificity	-1.6762930030826100	1E-06	-1676291.3267912800	0.0
high_specificity	~	frequency_of_transactions	8.881407684204260	1E-06	8881398.802805460	0.0
high_specificity	~	money	-6.0439824848173E-05	1E-06	-60.43976440840860	0.0
high_specificity	~	time	1.7726853023488900	9.70087956531196E-05	18273.4490428491	0.0
high_specificity	~	Foreign_Investment	-0.0057255700321304200	1E-06	-5725.564306566120	0.0
high_specificity	~	if_the_buyer_is_a_bank	0.0055864885746174600	3.11301821178311E-06	1794.556407963050	0.0
high_specificity	~	if_the_outcome_is_a_patent	-449.86842074592500	1E-06	-449867970.8779540	0.0
if_the_buyer_is_a_bank	~	frequency_of_transactions	7.4882581676853600	4.65679026427467E-07	16080264.865840700	0.0
if_the_outcome_is_a_patent	~	money	-0.0005887061644400470	3.51329917326873E-09	-167517.37271487900	0.0
Foreign_Investment	~	if_the_buyer_is_a_bank	-0.6939111245220970	1E-06	-693910.4306116660	0.0
Foreign_Investment	~~	Foreign_Investment	2.6816093613252300	0.010229972783109100	262.1325997553720	0.0
frequency_of_transactions	~~	frequency_of_transactions	157.3992905924530	9.39272789223643E-07	167575516.74315400	0.0
high_specificity	~~	high_specificity	0.04500000006276930	0.029054704689560600	1.5488025275778900	0.12142919777079800
if_the_buyer_is_a_bank	~~	if_the_buyer_is_a_bank	2.065327633658770	0.06787313437191690	30.42923614387980	0.0
if_the_outcome_is_a_patent	~~	if_the_outcome_is_a_patent	0.10633050067172600	0.0034942373284609000	30.430245757843600	0.0
money	~~	money	2147537721535704.5	1E-06	2.14753557400013E+21	0.0
time	~~	time	56486.4029484517	1E-06	56486346462.10520	0.0

Money: Has a significant negative impact on high specificity (Estimate = -676.26859, p-value = 0.0). This suggests that higher monetary values reduce the likelihood of high specificity, which might be counterintuitive but could imply that very high investments are associated with more general or versatile assets.

Time: Shows a positive impact on high specificity (Estimate = 0.00013387, p-value = 0.082). The effect is positive but not significant, indicating that time does not strongly influence high specificity.

Foreign Investment: Negatively influences high specificity (Estimate = -1.67629, p-value = 0.0), indicating that foreign investments are associated with lower specificity.

If the Buyer is a Bank: Positively influences high specificity (Estimate = 0.005586, p-value = 0.0), suggesting that transactions involving banks are more likely to be of high specificity.

If the Outcome is a Patent: Strongly negative impact on high specificity (Estimate = -449.86842, p-value = 0.0), indicating that patents are associated with higher specificity.

If the Buyer is a Bank ~ Frequency of Transactions: Positive relationship (Estimate = 7.48825, p-value = 0.0), indicating that higher frequency of transactions is associated with transactions involving banks.

If the Outcome is a Patent ~ Money: Negative relationship (Estimate = -0.0005887, p-value = 0.0), indicating that higher monetary values reduce the likelihood of the outcome being a patent.

Foreign Investment ~ If the Buyer is a Bank: Negative relationship (Estimate = -0.6939, p-value = 0.0), suggesting that foreign investments are less likely when the buyer is a bank.

Overall Model Fit:

Chi-square (χ^2): Indicates the difference between the observed and expected covariance matrices. The high value and low p-value suggest poor model fit.

CFI, GFI, AGFI, NFI, TLI: All show negative values, which typically indicates a poor model fit.

RMSEA: The value of 8.373521 indicates a poor fit (values >0.1 are considered poor).

AIC and BIC: Information criteria used for model comparison. Lower values indicate a better fit, but these values alone do not provide fit quality without comparison.

The SEM analysis for high specificity provides several interesting insights. Higher monetary values significantly reduce high specificity, suggesting that very high investments are associated with more versatile assets rather than highly specific ones. Foreign investments are associated with lower specificity. Higher frequency of transactions is associated with bank involvement but does not significantly impact high specificity directly. Outcomes resulting in patents are significantly associated with higher specificity.

Figure 25: Model Estimates for Frequency of Transactions

model_estimates_for_frequency_of_transactions

lval	op	rval	Estimate	Std. Err	z-value	p-value
frequency_of_transactions	~	high_specificity	1.0	-	-	-
frequency_of_transactions	~	low_specificity	1.0	-	-	-
money	~	high_specificity	0.006863598005785300	1.73799598369612E-13	5847333748.724000	0.0
money	~	low_specificity	0.0034086032747943200	1E-06	3408.59986619445	0.0
time	~	high_specificity	-2.70477131222173E-09	6.20787200796609E-07	-0.004356995362912680	0.9965236316673330
time	~	low_specificity	-3.52100141471463E-09	6.20792334638031E-07	-0.005671777092722230	0.9954746008883840
Foreign_Investment	~	high_specificity	3.27132584552117E-06	1E-06	3.2713225741986000	0.001070457342696290
Foreign_Investment	~	low_specificity	7.40760434000924E-07	1E-06	0.7407596932412310	0.458839158344162
high_specificity	~	frequency_of_transactions	0.0007546113993662830	7.0177141944038E-06	107.5294990551970	0.0
high_specificity	~	money	-3.5379355446832600	3.70738287688063E-06	-954294.365562179	0.0
high_specificity	~	time	0.005388491153570260	0.0006465908820292170	8.333694914357100	0.0
high_specificity	~	Foreign_Investment	-1.731133032547E-05	1.13285331979007E-09	-15267.698231615300	0.0
high_specificity	~	if_the_buyer_is_a_bank	1.654075170305E-05	1.62425062389197E-07	101.83557650628700	0.0
high_specificity	~	if_the_outcome_is_a_patent	-0.010367349639814900	1E-06	-10367.339272475600	0.0
low_specificity	~	frequency_of_transactions	0.0005386741901460460	7.01960663555773E-06	76.73850421744350	0.0
low_specificity	~	money	3.537893795189700	3.70921953045842E-06	953810.5826111260	0.0
low_specificity	~	time	0.005388491153636060	0.0006465909220077460	8.333694399188950	0.0
low_specificity	~	Foreign_Investment	-1.73113303192267E-05	2.29555697200366E-09	-7537.949430500400	0.0
low_specificity	~	if_the_buyer_is_a_bank	1.65407517030776E-05	1.62554071281667E-07	101.75475654288600	0.0
low_specificity	~	if_the_outcome_is_a_patent	-0.006912354845005950	4.79761637476887E-08	-144075.93748888400	0.0
if_the_buyer_is_a_bank	~	frequency_of_transactions	0.023746985764310100	6.19932940351043E-06	3830.5726939256400	0.0
if_the_outcome_is_a_patent	~	money	-0.0007224897800510770	1.73518486195694E-10	-4139904.0055901200	0.0
Foreign_Investment	~	if_the_buyer_is_a_bank	-0.0020765269005526200	1E-06	-2076.5248240278000	0.0
Foreign_Investment	~~	Foreign_Investment	0.09986690228350680	5.18541090865537E-09	19255493.643367800	0.0
frequency_of_transactions	~~	frequency_of_transactions	174.88728760547700	1.31890521936339E-10	1316025289518.0600	0.0
high_specificity	~~	high_specificity	0.04999950006927960	1.31882466763608E-10	376268602.5254670	0.0
if_the_buyer_is_a_bank	~~	if_the_buyer_is_a_bank	0.12810825935077100	1.89854897774912E-11	6410063539.951570	0.0
if_the_outcome_is_a_patent	~~	if_the_outcome_is_a_patent	0.11687425767840600	1.56101974263593E-06	74870.40657836030	0.0
low_specificity	~~	low_specificity	0.04999950006927960	1.31887812694723E-10	376253465.65182000	0.0
money	~~	money	2147537721535704.5	1E-06	2.14753557400013E+21	0.0
time	~~	time	56486.402944235	1E-06	56486346457.88850	0.0

The provided table and SEM results show the relationships between various predictors and the outcome variables of high_specificity and low_specificity.

Frequency of Transactions:

High Specificity: The estimate is set to 1.0 (standardized loading), indicating that it is a key predictor.

Low Specificity: The estimate is also 1.0, indicating it is a key predictor.

Money:

High Specificity: Direct effect (Estimate = 0.0068635980058, p-value = 0.0) indicates a positive but very small effect. Indirect effect via patent (Estimate = -0.010637464, p-value = 0.0) indicates a negative effect, implying more money reduces the likelihood of an outcome being a patent.

Low Specificity: Direct effect (Estimate = 0.003408603247, p-value = 0.0) indicates a small positive effect.

Time:

High Specificity: Direct effect (Estimate = -2.7047713, p-value = 0.997) indicates an insignificant effect.

Low Specificity: Direct effect (Estimate = -3.520107144, p-value = 0.995) indicates an insignificant effect.

Foreign Investment:

High Specificity: Direct effect (Estimate = 3.27132584552117, p-value = 0.0) indicates a significant positive effect. Indirect effect via bank involvement (Estimate = -1.731130302, p-value = 0.0) indicates a significant negative effect, suggesting foreign investment is less likely when the buyer is a bank.

Low Specificity: Direct effect (Estimate = 7.407604344000924, p-value = 0.0) indicates a significant positive effect.

If the Buyer is a Bank:

High Specificity: Direct effect (Estimate = 1.65407510730, p-value = 0.0) indicates a significant positive effect.

Low Specificity: Direct effect (Estimate = 1.65407510730, p-value = 0.0) indicates a significant positive effect.

Influence on Foreign Investment: Direct effect (Estimate = -0.69391112452, p-value = 0.0) indicates a significant negative effect.

If the Outcome is a Patent:

High Specificity: Direct effect (Estimate = -0.010637464, p-value = 0.0) indicates a significant negative effect.

Low Specificity: Direct effect (Estimate = -0.00619235484505, p-value = 0.0) indicates a significant negative effect.

Monetary Impact: The direct effect of money on both high and low specificity is positive but very small, indicating that while monetary investment influences specificity, its direct impact is minimal. Indirectly, more money reduces the likelihood of an outcome being a patent, which can influence specificity.

Time: Time has an insignificant direct effect on both high and low specificity, suggesting that the duration does not significantly impact asset specificity in the model.

Foreign Investment: Foreign investment significantly positively impacts both high and low specificity, indicating that foreign investment correlates with higher asset specificity. However, foreign investment is less likely when the buyer is a bank, indicating an inverse relationship between foreign investment and bank involvement.

Bank Involvement: Transactions involving banks significantly positively impact both high and low specificity, suggesting banks are associated with transactions of higher specificity. Bank involvement negatively influences foreign investment, indicating that banks might deter foreign investments.

Patent Outcomes: Patent outcomes significantly negatively impact both high and low specificity, indicating that outcomes involving patents are associated with lower specificity, potentially because patents might represent more generalizable innovations. Here 'Frequency_of_transactions' is a critical predictor for both high and low specificity, serving as a baseline for the model. While direct impacts of money and foreign investment are minimal, their indirect pathways provide deeper insights into how financial aspects influence specificity. If the Buyer is a Bank positively correlates with specificity but negatively impacts foreign investments, highlighting a complex interplay between bank involvement and investment sources. Outcomes resulting in patents negatively impact specificity, suggesting that more general innovations reduce asset specificity.

7.6 Insights From Analysis

By combining insights from Causal ML and SEM analyses, we gain a comprehensive understanding of factors influencing asset specificity. The Causal ML analysis reveals that financial metrics significantly drive high specificity, with higher investments correlating with increased specificity, and banks positively influencing high specificity but negatively impacting low specificity. R&D missions are linked to higher specificity, while export activities show variable effects. The SEM analysis highlights that transaction frequency, particularly involving banks, is crucial for both high and low specificity. Monetary investments have minimal direct effects but complex indirect relationships, especially reducing patent outcomes. Time duration of transactions is insignificant, whereas foreign investments significantly impact specificity but are less likely when banks are involved. Patent outcomes negatively affect specificity,

indicating more general innovations, and R&D missions are associated with higher specificity. Both analyses agree that monetary investments, transaction frequency, and bank involvement are key predictors, with foreign investments playing a significant role but being deterred by bank transactions. Patents consistently show a negative impact on specificity, while R&D missions and export activities have nuanced effects. This integrated approach underscores the multifaceted nature of asset specificity, influenced by financial, transactional, and institutional factors.

Chapter 8 Discussion

8.1 Interpretation of Findings

The findings from the machine learning analysis provide a nuanced understanding of the factors that contribute to the classification of economic transactions. The Random Forest Classifier, optimized through rigorous hyperparameter tuning and validated using Stratified K-Fold cross-validation, has demonstrated a high level of accuracy, evidenced by near-perfect precision, recall, and F1-scores. The model's performance, bolstered by an average cross-validation score of 0.884, signifies its robustness and the generalizability of its predictions.

The pivotal role of the 'frequency of transactions' feature, as revealed by the feature importance analysis, underscores the centrality of transaction size in economic activity categorization. This aligns with economic theory, which posits that the size of a transaction can be reflective of its underlying complexity and the degree of asset specificity involved. Asset specificity refers to the extent to which the assets involved in a transaction are tailored to the transaction. A higher degree of specificity can often lead to more significant investments in transaction-specific assets, thereby increasing the importance of the transaction's frequency of transactions.

The model's sensitivity to the 'frequency of transactions' feature further suggests that as the size of a transaction increases, the economic actors likely engage in more detailed and binding contracts to safeguard their investments. This observation is particularly relevant in the analysis of asset specificity in economic transactions. High asset specificity transactions often require a greater commitment from the parties involved, which may influence the transaction's categorization within our model.

Moreover, the importance attributed to features related to 'Turnover' and 'money' indicates that the financial characteristics of the parties involved in the transaction are also key determinants. This could be interpreted as a reflection of the economic principle that transactions are not only defined by the exchange of goods or services but also by the financial health and liquidity of the entities engaged in the exchange.

The steep drop in importance scores beyond these top features suggests a diminishing marginal utility of additional data for improving model predictions. This could indicate that once the model has access to the core financial characteristics of a transaction, additional details contribute less to the classification decision. This finding has practical implications for economic research and business practice, as it may guide the focus of data collection and analysis on those attributes most impactful for transaction categorization. The Random Forest Classifier has not only served as a predictive model but also as a tool for uncovering the underlying structure of economic transactions. The significant features identified by the model offer insights into the critical factors that define and influence the categorization of transactions, emphasizing the role of transaction size and the financial attributes of the entities involved. These insights contribute to a more profound understanding of asset specificity in economic

transactions, with potential applications in economic theory, financial analysis, and business strategy development.

8.2 Practical Implications for Businesses and Policymakers

The application of causal ML analysis and RFC in evaluating asset specificity provides valuable insights that can guide government policy and stimulate investment from MNCs. By leveraging these advanced analytical techniques, policymakers can make informed decisions that promote high-specificity (HS) work and create a conducive environment for foreign direct investment (FDI). The following recommendations outline how these insights can be operationalized:

Targeted Incentives for Specialized Investments: The analysis indicates that higher financial investments are often associated with more specialized, less substitutable assets. Governments can use this information to design targeted incentives, such as tax breaks or grants, for companies investing in high-specificity projects. This could include subsidies for R&D activities, especially those that involve significant customization or innovation.

Enhancing Infrastructure for Specialized Sectors: Insights from the temporal and financial dimensions of asset specificity highlight the importance of robust infrastructure. Governments can invest in specialized industrial zones or technology parks that provide state-of-the-art facilities and services tailored to the needs of high-specificity sectors. This would reduce setup costs and risks for companies, making high-specificity investments more attractive.

Streamlining Regulatory Frameworks: The importance of transaction characteristics such as buyer and seller classifications suggests that regulatory simplicity and clarity are crucial. Simplified regulations and expedited approval processes for projects involving high-specificity assets can reduce bureaucratic delays and costs, encouraging more firms to engage in such work.

Leveraging Technology Transfer Policies: According to research on China's 'Market for Technology' policy, active involvement of MNCs in technology transfer can significantly enhance domestic technological capabilities. The government can facilitate partnerships between domestic firms and MNCs by offering co-investment opportunities and collaborative research initiatives that align with the identified high-specificity areas.

Creating a Stable and Predictable Investment Climate: The analysis of transaction data underscores the need for a stable investment environment. Governments can foster this by ensuring policy stability, protecting intellectual property rights, and maintaining transparent and consistent enforcement of laws. This predictability is crucial for MNCs when making long-term, high-specificity investments.

Promoting Open Innovation Frameworks: The outbound open innovation model shows that MNCs benefit from and contribute to open innovation ecosystems. Policies that encourage open innovation, such as shared R&D facilities, innovation clusters, and platforms for knowledge exchange, can attract MNCs to invest in local high-specificity projects. These platforms can also facilitate the flow of expertise and technology between local and international firms.

Data-Driven Decision Making: The feature importance scores from the RFC analysis highlight key factors influencing asset specificity. Governments can use these data-driven insights to prioritize sectors and projects for investment and support. For instance, focusing on sectors where high transaction amounts correlate with significant asset specificity can maximize economic impact.

Real-Time Monitoring and Adaptation: Machine learning models can provide real-time analysis of ongoing projects, allowing for dynamic policy adjustments. By continuously monitoring the performance and outcomes of high-specificity investments, governments can quickly identify and address challenges, ensuring that policies remain effective and responsive to changing conditions.

8.3 Limitations and Future Research Directions

This study has provided valuable insights into the categorization of economic transactions through machine learning. However, it is important to acknowledge its limitations and the opportunities they present for future research.

One primary limitation is the reliance on a single algorithm, the Random Forest Classifier. While this model has demonstrated robust performance, different algorithms may reveal additional nuances in the data. Future research could involve comparative analyses using other machine learning techniques, such as support vector machines or neural networks, to validate and potentially expand upon the findings of this study. Another limitation arises from the feature set used. Further research should consider exploring a broader set of features, including macroeconomic indicators or more granular transaction-level details. The model's interpretability is also a concern. Random Forests, although powerful, do not readily lend themselves to interpretability in the same way that simpler models like decision trees do. The development of models that maintain high accuracy while also providing more interpretable results would be a valuable direction for future research. Additionally, the dataset used in this study was limited to a particular period and context. The generalizability of the findings may be restricted by these conditions. To address this, future studies should aim to replicate the analysis across different datasets, timeframes, and economic environments to assess the consistency of the model's performance and the generalizability of the results.

The near-perfect classification metrics could suggest an overfitting issue or an underrepresentation of the complexity inherent in real-world data. This could be explored further by applying the model to out-of-sample data and assessing its

performance in a real-world scenario. It would also be prudent to investigate the model's resilience to shifts in economic conditions and transaction dynamics.

Moreover, the potential impact of evolving financial regulations and emerging market trends on transaction categorization could be an area ripe for exploration. As markets evolve and new types of transactions emerge, it is crucial to continually refine and adapt predictive models to maintain their accuracy and relevance. While this study has laid a foundation for the application of machine learning in understanding economic transactions, it also opens several pathways for further investigation. By addressing these limitations, future research can continue to advance the intersection of machine learning and economic analysis, ultimately contributing to more informed business and policy decisions.

Chapter 9 Conclusion

9.1 Summary of Key Findings

This thesis has explored the application of machine learning, specifically through the use of an optimized Random Forest Classifier, to categorize economic transactions based on various features. The analysis was grounded in a comprehensive dataset, processed and analyzed to discern patterns and predictors of transaction categorization. Here, we summarize the pivotal findings:

Significance of Transaction Size: The 'frequency of transactions' feature, representing the size of the transactions, emerged as the most significant predictor of transaction categorization. This finding underscores the critical role of transaction size in determining the nature and complexity of economic exchanges. It suggests that larger transactions often involve more complex considerations and contractual arrangements, reflecting their heightened importance in economic activity.

Financial Attributes as Key Predictors: Besides transaction size, financial attributes such as 'Turnover' and 'money' were identified as crucial in influencing the categorization of transactions. These features highlight the significance of the financial health and liquidity of the entities involved in transactions, pointing to the broader economic context within which these transactions occur.

Efficacy of the Random Forest Classifier: The optimized Random Forest Classifier demonstrated high accuracy, precision, recall, and F1-scores, indicating its robustness in modeling and predicting transaction categories. This outcome validates the suitability of Random Forest as a powerful tool for classifying complex economic data, benefiting from its ability to handle high-dimensional datasets and capture non-linear relationships.

Impact of Feature Binning on Model Performance: The application of binning to the 'frequency of transactions' feature enhanced the model's performance, suggesting that discretizing continuous variables into meaningful categories can improve predictive accuracy. This technique not only simplified the model's input space but also facilitated a more nuanced interpretation of the data, allowing for clearer insights into the role of transaction size.

Stratified K-Fold Cross-Validation for Model Validation: Employing Stratified K-Fold cross-validation ensured that the model's evaluation was robust and reflective of the dataset's inherent class distribution. This approach verified the model's generalizability and reliability across different subsets of the data, confirming its applicability to diverse economic transactions.

Practical Implications for Economic Research and Business Practices: The findings have significant practical implications, suggesting that businesses and policymakers can leverage such predictive modeling to refine risk assessment, regulatory compliance, and strategic decision-making processes. By focusing on key transaction attributes, stakeholders can better allocate resources, manage risks, and foster economic stability.

Causal ML Findings: The integration of Causal ML analysis provided deeper insights into the causal relationships between various features and asset specificity. Financial metrics were found to significantly drive high specificity, with higher investments correlating with increased specificity. Banks positively influenced high specificity but negatively impacted low specificity. R&D missions were linked to higher specificity, while export activities exhibited variable effects. This causal analysis underscores the importance of financial and institutional factors in shaping asset specificity.

SEM Analysis Findings: Structural Equation Modeling (SEM) highlighted the critical role of transaction frequency, particularly involving banks, in influencing both high and low specificity. Monetary investments showed minimal direct effects but had complex indirect relationships, particularly reducing patent outcomes. Time duration of transactions was found to be insignificant, whereas foreign investments significantly impacted specificity but were less likely when banks were involved. Patent outcomes consistently showed a negative impact on specificity, while R&D missions and export activities had nuanced effects. This integrated approach from SEM and Causal ML analyses provided a comprehensive understanding of the multifaceted nature of asset specificity.

Practical Implications for Economic Research and Business Practices: The findings have significant practical implications, suggesting that businesses and policymakers can leverage such predictive modeling and causal analysis to refine risk assessment, regulatory compliance, and strategic decision-making processes. By focusing on key transaction attributes, stakeholders can better allocate resources, manage risks, and foster economic stability. The integration of Causal ML and SEM analyses further enhances the ability to make informed decisions based on a comprehensive understanding of the underlying factors influencing asset specificity. This study demonstrates the value of combining advanced ML techniques with traditional economic theories, suggesting a paradigm shift towards more empirical and quantitative analyses in economics.

9.2 Final Thoughts

As this thesis journey concludes, it is essential to step back and reflect on the broader narrative that has unfolded through the exploration of machine learning applications in categorizing economic transactions. This research has not only contributed to the academic discourse on the intersection of economics and technology but has also illuminated pathways for practical implementation in business and policy-making spheres. Here, I offer some final reflections on the significance of this work, its implications, and the personal journey that accompanied its realization.

This thesis has underscored the transformative potential of machine learning in deciphering complex economic phenomena. By harnessing the power of an optimized Random Forest Classifier, we have peeled back layers of transactional data to reveal the intricate dance of variables that influence transaction categorization. This endeavor, rooted in empirical analysis, offers a beacon for future explorations into the vast seas of economic data, where untold stories await discovery.

One of the most gratifying aspects of this research has been its ability to bridge theoretical constructs with practical applications. The insights gleaned from the feature importance analysis and model performance evaluation illuminate a path for businesses and policymakers to navigate the complexities of economic transactions more adeptly. This confluence of theory and practice embodies the essence of applied research, where the fruits of academic labor can directly inform and enhance real-world decision-making processes. Embarking on this research journey has been both challenging and immensely rewarding. The process of transforming a curiosity-driven question into a structured investigation, grappling with data, wrestling with models, and finally distilling insights, has been a profound learning experience. It has underscored the importance of perseverance, critical thinking, and the willingness to venture into uncharted territories. The challenges encountered along the way, from data preprocessing hurdles to model optimization conundrums, have been invaluable lessons in problem-solving and resilience. While this thesis marks the culmination of one research endeavor, it also opens the door to myriad future investigations. The findings presented here are but a snapshot in the continuously evolving landscape of economic analysis and machine learning applications. There is vast potential for further research to expand on the foundations laid by this work, exploring new algorithms, datasets, and methodologies. The digital economy continues to evolve, the importance of understanding and categorizing economic transactions will only grow. Machine learning offers a powerful toolkit for navigating this complexity, and it is my hope that this thesis will inspire others to explore these tools further, pushing the boundaries of what is possible in economic analysis.

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Acknowledgments

I would like to express my heartfelt gratitude to my advisor, Professor Zhang Fei Hu, for his invaluable guidance throughout my research journey. His expertise and patient instruction provided the foundation for my work.

My sincere appreciation also goes to the dedicated research team working under Professor Zhang. Their willingness to share their knowledge and insights greatly enhanced my understanding of the complex data involved in this study.

I am equally thankful for the support of my friends, whose proficiency in the Chinese language was instrumental in helping me navigate and comprehend the governmental data critical to my thesis.

Their collective contributions have been a cornerstone of my academic endeavor, and for that, I am profoundly grateful.